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CHILDREN'S BODY MEASUREMENTS

FOR SIZING GARMENTS AND PATTERNS

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UNITED STATES DEPARTMENT OF AGRICULTURE

MISCELLANEOUS PUBLICATION NO. 365

Children's Body Measurements for Sizing Garments and Patterns

A proposed standard system based on height and girth of hips

by Ruth O'Brien, *chief*
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Foreword

The proposed standard system of children's body measurements suggested here is the result of many months' study by statisticians of 4,917,052 body measurements taken on 133,807 American boys and girls as part of a cooperative Works Progress Administration research project, initiated and led by the Bureau of Home Economics of the United States Department of Agriculture. A complete technical report of the study is now in preparation.

Eighteen colleges, universities, and other educational institutions helped in this research by giving the part-time services of one of their teachers who arranged local training schools for measurers, obtained permission to measure children in various counties in their States, and assisted in supervision.

Acknowledgment is here given to these cooperators:

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Appreciation is also expressed to Lennah Curtiss Zens, administrative assistant; to Eleanor P. Hunt, the anthropometrist in charge of the measuring; to her assistants Charles E. Snow and Everett L. Marshall; to Catherine E. Hoffmann, who supervised the measuring in Maryland and the District of Columbia; and to Albert E. Craig and Cassie F. Skilling, who assisted in the analysis of the data.

LOUISE STANLEY,
Chief, Bureau of Home Economics.

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Children's Body Measurements for Sizing Garments and Patterns

Need for Scientific Measurements

The rapid growth of the pattern and ready-to-wear industry in the United States has brought with it many difficulties in the proper sizing of garments and patterns, especially those manufactured for women and children. Any satisfactory American sizing system must be based on dimensions obtained by measuring large numbers of persons throughout the country. Each measurement must be made in exactly the same way by individuals carefully trained in a method that can be duplicated. The same kind of instruments must be used, and these must be constantly checked so that they are accurate.

Unfortunately no such large, scientific study of the body measurements used in the construction of women's and children's garments has ever been reported. When funds were made available by the Works Progress Administration in 1937, the Bureau of Home Economics, therefore, organized and directed such a cooperative research project. The measurements used in the construction of trunk garments were taken on 147,088 children, 4 to 17 years of age, inclusive, distributed in 15 States and the District of Columbia.

This proposed standard of body measurements is based on the results of that study. It recommends dimensions to be used in constructing a series of standard mannequins such as are used by manufacturers to size garments and patterns. The measurements were taken next to the skin. The proposed standard, therefore, does not give garment and pattern dimensions. Standards for these can be developed from this proposed basis by agreement in the trade on tolerances for construction, style, and other clothing features.

Measurements Included in the Study

The 36 measurements made on each child were chosen after consultation with retailers and garment and pattern manufacturers. These measurements are listed in the appendix (p. 19), together with a description of the method used in taking each. The boys wore trunks and the girls trunks and bandeaux. Men measured boys and women measured girls. Measurers were trained in the particular methods they were to use, and every possible precaution was taken to see that all measurements were made accurately and in the same way with reliable, calibrated instruments.

The first plan was to include measurements of feet, hands, and heads as the basis for sizing shoes, gloves, and hats. This idea had to be abandoned, however. Trials showed that such a large number of measurements fatigued the children too much. The results were inaccurate, and school authorities were unwilling to grant permission to measure a large group of children more than once. The list was therefore restricted to weight and the measurements used in making garments worn on the trunk of the body. The age range included in the study was limited by the funds available. The ages 4 to 17 were chosen because large groups of children of these ages could be reached in the schools, thus reducing the time and expense involved.

Number of Children Measured

Although 147,088 children were measured, the size system proposed here is based on measurements obtained on 133,807 children (69,661 boys

and 64,146 girls) which was the number of complete records available when this part of the study was started. The distribution by States is shown in table 1. Some parts of the country are not represented. The taking of measurements depended on the cooperation of local institutions that would act as sponsors, and on school boards and other authorities willing to allow children to be measured. Unfortunately, such cooperation could not be secured in all sections of the country.

TABLE 1.—*Number of children by States*

State	Boys	Girls	Total
Alabama	4,106	4,446	8,552
California	2,625	2,484	5,109
Colorado	4,303	4,302	8,605
District of Columbia and Maryland	6,521	5,244	11,765
Illinois	6,978	6,581	13,559
Iowa	3,961	4,010	7,971
Kansas	3,513	4,240	7,753
Michigan	6,846	3,754	10,600
Minnesota	5,672	5,969	11,641
Nebraska	3,510	3,539	7,049
Ohio	7,808	3,774	11,582
Pennsylvania	3,249	4,198	7,447
Tennessee	2,918	2,899	5,817
Texas	3,514	4,046	7,560
Utah	4,137	4,660	8,797
Total	69,661	64,146	133,807

Age, the Poorest Predictor of Body Dimensions

Essentially the problem of sizing garments is one of finding that measurement or combination of measurements which best predicts the other dimensions of a child's body. This is especially true when what is required is the creation of a representative form or model upon which standard garments may be manufactured. It follows, therefore, that the best choice is that measurement or combination of measurements which is most closely related to the greatest number of the others, provided, of course, it also satisfies the further criterion of being practicable.

Age is now used as a basis for sizing practically all children's garments and patterns, and many persons have surmised that this is the underlying cause of size difficulties. A great many children of exactly the same age have entirely different dimensions and body proportions. The first step in this study was therefore to check this point and to determine by a statistical analysis of the measurements obtained what basis is best.

Best size basis sought

Briefly, three independent samples from the records were used for this analysis. Two of these included approximately 8,000 children each, and one 16,000. Each of these three samples gave similar results. The analysis was carried out separately for boys and for girls. Eighteen important body measurements in addition to age and weight were selected for study. The correlation coefficient was calculated for the 190 possible pairs of the 20 items. In general terms the correlation coefficient is an index which measures the closeness with which one measurement can be predicted from another. The correlation coefficient never exceeds unity; and the nearer to one it is, the more highly the two measurements are related.

Tables 2 and 3 were obtained by combining the three samples upon which the analysis was carried out separately. Table 2 gives the correlations of each of the 20 items with every other item for a sample of 32,165 boys, aged 4 to 14, and table 3 gives similar correlations for 31,919 girls, aged 4 to 14. For example, column 2 in these tables gives the correlation of age (in months) with every other measurement, and column 3 gives the correlation of waist height with every other measurement.

Note that on the whole the set of correlations in column 2 is smaller than the sets in any other column. Thus, of the entire set, age is the least highly correlated with all the other measurements. Note also that on the whole, lengths are more highly correlated with other lengths than with girths and that girths are more highly correlated with other girths than with lengths. It should be pointed out, however, that from a statistical point of view the close relationship found between two lengths is not so significant when one of the lengths forms part of the other as when they are independent of each other. An example of the first would be stature and hip height, and of the second stature and length of arm.

TABLE 2.—*Intercorrelations of 20 measurements on a sample of 32,165 boys aged 4–14*

	Age	Waist height	Hip height	Weight	Stature	Cervicale height	Tibiale height	Bitrochanteric diameter	Chest girth	Waist girth	Hip girth	Neck-base girth	Armscye girth	Upper-arm girth	Posterior arm length	Thigh girth	Maximum calf girth	Knee girth	Crotch length	Vertical trunk girth
Age		0.899	0.900	0.822	0.897	0.898	0.879	0.826	0.813	0.690	0.807	0.760	0.823	0.688	0.887	0.736	0.785	0.812	0.799	0.848
Waist height	0.899		.991	.915	.990	.991	.977	.909	.886	.777	.896	.841	.891	.766	.974	.824	.864	.901	.904	.933
Hip height	.900	.991		.901	.985	.986	.978	.894	.875	.764	.881	.828	.881	.750	.973	.808	.852	.890	.873	.916
Weight	.822	.915	.901		.927	.926	.897	.961	.958	.903	.975	.888	.943	.912	.914	.937	.948	.947	.933	.961
Stature	.897	.990	.985	.927		.996	.972	.918	.896	.782	.904	.857	.901	.776	.973	.828	.875	.907	.903	.951
Cervicale height	.898	.991	.986	.926	.996		.973	.918	.895	.783	.904	.851	.900	.775	.974	.828	.873	.906	.901	.949
Tibiale height	.879	.977	.978	.897	.972	.973		.890	.870	.767	.877	.824	.872	.750	.960	.806	.852	.900	.866	.909
Bitrochanteric diam.	.826	.909	.894	.961	.918	.918	.890		.931	.875	.973	.861	.921	.880	.907	.921	.928	.935	.914	.938
Chest girth	.813	.886	.875	.958	.896	.895	.870	.931		.900	.946	.879	.930	.899	.891	.911	.917	.918	.898	.932
Waist girth	.690	.777	.764	.903	.782	.783	.767	.875	.900		.909	.807	.856	.879	.782	.900	.870	.868	.836	.849
Hip girth	.807	.896	.881	.975	.904	.904	.877	.973	.946	.909		.868	.932	.922	.893	.964	.947	.952	.930	.944
Neck-base girth	.760	.841	.828	.888	.857	.851	.824	.861	.879	.807	.868		.878	.810	.845	.821	.846	.850	.849	.885
Armscye girth	.823	.891	.881	.943	.901	.900	.872	.921	.930	.856	.932	.878		.889	.898	.894	.902	.908	.891	.930
Upper-arm girth	.688	.766	.750	.912	.776	.775	.750	.880	.899	.879	.922	.810	.889		.771	.936	.900	.877	.845	.855
Posterior arm length	.887	.974	.973	.914	.973	.974	.960	.907	.891	.782	.893	.845	.898	.771		.821	.861	.895	.882	.924
Thigh girth	.736	.824	.808	.937	.828	.828	.806	.921	.911	.900	.964	.821	.894	.936	.821		.931	.917	.882	.886
Maximum calf girth	.785	.864	.852	.948	.875	.873	.852	.928	.917	.870	.947	.846	.902	.900	.861	.931		.936	.892	.912
Knee girth	.812	.901	.890	.947	.907	.906	.900	.935	.918	.868	.952	.850	.908	.877	.895	.917	.936		.903	.924
Crotch length	.799	.904	.873	.933	.903	.901	.866	.914	.898	.836	.930	.849	.891	.845	.882	.882	.892	.903		.942
Vertical trunk girth	.848	.933	.916	.961	.951	.949	.909	.938	.932	.849	.944	.885	.930	.855	.924	.886	.912	.924	.942	

TABLE 3.—*Intercorrelations of 20 measurements on a sample of 31,919 girls aged 4–14*

	Age	Waist height	Hip height	Weight	Stature	Cervicale height	Tibiale height	Bitrochanteric diameter	Chest girth	Waist girth	Hip girth	Neck-base girth	Armscye girth	Upper-arm girth	Posterior arm length	Thigh girth	Maximum calf girth	Knee girth	Crotch length	Vertical trunk girth
Age		0.915	0.908	0.843	0.921	0.921	0.890	0.857	0.826	0.673	0.840	0.799	0.823	0.692	0.909	0.765	0.800	0.823	0.826	0.885
Waist height	0.915		.989	.906	.990	.991	.974	.904	.875	.746	.895	.852	.877	.751	.975	.828	.854	.890	.901	.937
Hip height	.908	.989		.884	.982	.982	.974	.881	.856	.727	.871	.833	.856	.724	.971	.804	.834	.872	.862	.912
Weight	.843	.906	.884		.915	.916	.878	.967	.961	.889	.983	.889	.939	.910	.904	.955	.948	.948	.936	.960
Stature	.921	.990	.982	.915		.996	.968	.915	.881	.744	.902	.852	.881	.752	.974	.830	.862	.893	.895	.952
Cervicale height	.921	.991	.982	.916	.996		.968	.916	.882	.747	.904	.856	.882	.755	.974	.832	.862	.892	.896	.951
Tibiale height	.890	.974	.974	.878	.968	.968		.876	.850	.728	.866	.827	.849	.726	.957	.803	.833	.878	.854	.902
Bitrochanteric diam.	.857	.904	.881	.967	.915	.916	.876		.937	.852	.979	.865	.916	.872	.902	.931	.920	.931	.920	.947
Chest girth	.826	.875	.856	.961	.881	.882	.850	.937		.895	.953	.871	.927	.897	.877	.929	.916	.918	.903	.928
Waist girth	.673	.746	.727	.889	.744	.747	.728	.852	.895		.886	.787	.850	.881	.747	.896	.855	.854	.826	.823
Hip girth	.840	.895	.871	.983	.902	.904	.866	.979	.953	.886		.875	.931	.913	.892	.967	.944	.949	.937	.952
Neck-base girth	.799	.852	.833	.889	.852	.856	.827	.865	.871	.787	.875		.862	.795	.849	.839	.846	.857	.852	.881
Armscye girth	.823	.877	.856	.939	.881	.882	.849	.916	.927	.850	.931	.862		.885	.880	.907	.896	.903	.893	.920
Upper-arm girth	.692	.751	.724	.910	.752	.755	.726	.872	.897	.881	.913	.795	.885		.751	.939	.899	.880	.850	.842
Posterior arm length	.909	.975	.971	.904	.974	.974	.957	.902	.877	.747	.892	.849	.880	.751		.824	.850	.883	.875	.925
Thigh girth	.765	.828	.804	.955	.830	.832	.803	.931	.929	.896	.967	.839	.907	.939	.824		.937	.929	.903	.901
Maximum calf girth	.800	.854	.834	.948	.862	.862	.833	.920	.916	.855	.944	.846	.896	.899	.850	.937		.937	.892	.908
Knee girth	.823	.890	.872	.948	.893	.892	.878	.931	.918	.854	.949	.857	.903	.880	.883	.929	.937		.906	.921
Crotch length	.826	.901	.862	.936	.895	.896	.854	.920	.903	.826	.937	.852	.893	.850	.875	.903	.892	.906		.945
Vertical trunk girth	.885	.937	.912	.960	.952	.951	.902	.947	.928	.823	.953	.881	.920	.842	.925	.901	.908	.921	.945	

In order to put age on a relative basis with other measurements, it was taken in combination with each one of seven important measurements and its percentage contribution to the prediction of the other 18 measurements calculated. Table 4 gives the results obtained on the basis of age and height (stature), and age and girth of hips. The percentage contribution of age in the prediction of all the other measurements is almost negligible as compared with that of stature. In predicting other

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TABLE 4.—*Percentage contribution of age in predicting body measurements in contrast to the contribution of stature or of hip girth*

Measurements to be predicted	Boys				Girls			
	Age versus stature		Age versus hip girth		Age versus stature		Age versus hip girth	
	Age	Stature	Age	Hip girth	Age	Stature	Age	Hip girth
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*	*Percent*
Waist height	5.2	94.8	50.7	49.3	2.2	97.8	56.5	43.5
Hip height	8.5	91.5	55.1	44.9	2.4	97.6	61.8	38.2
Weight	4.8	95.2	10.4	89.6	.4	99.6	5.8	94.2
Stature			48.2	51.8			55.9	44.1
Cervicale height	2.2	97.8	48.5	51.5	2.2	97.8	55.3	44.7
Tibiale height	3.3	96.7	50.3	49.7	.8	99.2	57.7	42.3
Bitrochanteric diameter	1.1	98.9	11.8	88.2	10.4	89.6	11.6	88.4
Chest girth	5.5	94.5	14.6	85.4	10.9	89.1	9.1	90.9
Waist girth	6.9	93.1	11.0	89.0	8.5	91.5	18.3	81.7
Hip girth	2.7	97.3			7.0	93.0		
Neck-base girth	5.0	95.0	18.8	81.2	10.7	89.3	23.8	76.2
Armscye girth	7.8	92.2	20.9	79.1	8.3	91.7	14.3	85.7
Upper-arm girth	5.1	94.9	13.2	86.8	.5	99.5	18.5	81.5
Posterior arm length	7.3	92.7	48.4	51.6	8.3	91.7	55.6	44.4
Thigh girth	4.1	95.9	10.1	89.9	.8	99.2	12.8	87.2
Maximum calf girth	.1	99.9	6.4	93.6	5.1	94.9	2.3	97.7
Knee girth	1.1	98.9	13.1	86.9	1.0	99.0	9.0	91.0
Crotch length	5.5	94.5	14.7	85.3	1.4	98.6	13.5	86.5
Vertical trunk girth	2.4	97.6	25.1	74.9	5.7	94.3	28.8	71.2

girths. age contributes much less than hip girth. This is true for both boys and girls. This fact, when taken in conjunction with the results of tables 2 and 3, shows that a system of sizing much superior to that of age can be found.

A study of a more technical nature showed that if only one measurement were to be used as a basis for garment and pattern sizes, weight is the best one, girth of hips is second best, and age is the most unreliable.

Combination of a Body Length and a Girth Needed

Though weight or girth of hips, alone, is a fairly good measurement for sizing, neither is sufficient by itself to explain the variation of all the other measurements, especially the lengths. This is apparent from figures 1 and 2 which show the great variation in height for a given hip girth.

An analysis was next undertaken, therefore, to determine which combination of two measurements would best predict the other measurements of the set. To this end, 8 of the 20 measurements were chosen which in combination seemed likely to be good measurements for the purpose of sizing garments. The 8 items chosen were: Waist height, weight, stature (i. e., height), girth of chest at the armscye, girth of hips, total posterior length of the arm, and the vertical girth of the trunk, age being included as a further check.

These 8 measurements were divided into every possible combination of 2, and regression equations and multiple correlations were calculated for each of the 20 items on each of the combinations. In general, the regression equation is a formula which gives the predicted value of one measurement from the known values of a combination of others. The multiple correlation coefficient, like the simple correlation coefficients given in tables 2 and 3, is an index which measures how closely a measurement can be predicted from any given combination of other measurements.

This study has brought out several important facts. It showed, for instance, that if the measurements are divided into two categories, namely, lengths and girths, then in predicting a length, another length is the most important factor, while in predicting a girth, another girth is most important. Tables 5 and 6 illustrate this point well. Columns 2, 3, 7, and 8 of table 5 give the coefficient of the regression equation of 17 measurements on stature and girth of hips, and

columns 4, 5, 9, and 10 give the percentage each coefficient is of the sum of both coefficients. These coefficients in standard units express the relative importance of a particular measurement in predicting the other measurements.

Thus, in predicting arm length on the basis of stature and girth of hips (table 5), in the case of the boys, stature contributes 92.6 percent, while girth of hips contributes only 7.4 percent. On the other hand, in predicting girth of chest, stature contributes only 22.5 percent, while girth of hips contributes 77.5 percent. It is also interesting to note that in predicting weight, the girth of hips contributes much more than stature.

The same is true in table 6, where the girth of chest was included in the regression equation for the prediction of the other measurements. This, together with a more extensive analysis, shows that the best predictor of both girths and lengths is a girth (or weight) in combination with a length.

TABLE 5.—*Standard coefficients of regression, with corresponding percentage contribution and multiple correlation, of 17 measurements on stature and hip girth*

Measurements	Boys					Girls				
	Standard coefficient of—		Percentage contribution of—		Multiple correlation	Standard coefficient of—		Percentage contribution of—		Multiple correlation
	Stature	Hip girth	Stature	Hip girth		Stature	Hip girth	Stature	Hip girth	
Waist height	0.9885	0.0021	99.8	0.2	0.990	0.9835	0.0075	99.2	0.8	0.990
Hip height	1.0335	−.0539	95.0	5.0	.985	1.0522	−.0781	93.1	6.9	.982
Weight	.2495	.7490	25.0	75.0	.980	.1502	.8474	15.1	84.9	.985
Cervicale height	.9803	.0175	98.3	1.7	.996	.9714	.0274	97.3	2.7	.996
Tibiale height	.9812	−.0101	99.0	1.0	.972	.9998	−.0355	96.6	3.4	.968
Bitrochanteric diameter	.2098	.7830	21.1	78.9	.977	.1689	.8265	17.0	83.0	.982
Chest girth	.2175	.7497	22.5	77.5	.951	.1179	.8464	12.2	87.8	.954
Waist girth	−.2185	1.1065	16.5	83.5	.914	−.3009	1.1578	20.6	79.4	.896
Neck-base girth	.3928	.5132	43.4	56.6	.884	.3388	.5694	37.3	62.7	.887
Armscye girth	.3218	.6408	33.4	66.6	.942	.2215	.7315	23.2	76.8	.936
Upper-arm girth	−.3176	1.2091	20.8	79.2	.932	−.3819	1.2572	23.3	76.7	.927
Posterior-arm length	.9075	.0728	92.6	7.4	.974	.9106	.0700	92.9	7.1	.974
Thigh girth	−.2395	1.1803	16.9	83.1	.969	−.2295	1.1741	16.4	83.6	.972
Maximum calf girth	.1024	.8543	10.7	89.3	.948	.0516	.8978	5.4	94.6	.945
Knee girth	.2572	.7188	26.4	73.6	.958	.1944	.7739	20.1	79.9	.953
Crotch length	.3387	.6235	35.2	64.8	.941	.2634	.6996	27.4	72.6	.944
Vertical trunk girth	.5329	.4619	53.6	46.4	.971	.4975	.5033	49.7	50.3	.976

TABLE 6.—*Standard coefficients of regression, with corresponding percentage contribution and multiple correlation, of 16 measurements on stature, chest girth, and hip girth*

Measurements	Boys							Girls						
	Standard coefficient of—			Percentage contribution of—			Multiple correlation	Standard coefficient of—			Percentage contribution of—			Multiple correlation
	Stature	Chest girth	Hip girth	Stature	Chest girth	Hip girth		Stature	Chest girth	Hip girth	Stature	Chest girth	Hip girth	
Waist height	0.9925	−0.0183	0.0158	96.7	1.8	1.5	0.990	0.9819	.0131	−0.0037	98.3	1.3	0.4	0.990
Hip height	1.0315	.0093	−.0608	93.7	.8	5.5	.985	1.0478	.0375	−.1099	87.7	3.1	9.2	.983
Weight	.1907	.2701	.5466	18.9	26.8	54.3	.984	.1219	.2409	.6436	12.1	23.9	64.0	.988
Cervicale height	.9800	.0014	.0164	98.2	.2	1.6	.996	.9712	.0018	.0258	97.2	.2	2.6	.996
Tibiale height	.9786	.0119	−.0190	96.9	1.2	1.9	.972	.9966	.0266	−.0580	92.2	2.5	5.3	.968
Bitrochanteric diameter	.2057	.0189	.7688	20.7	1.9	77.4	.977	.1686	.0025	.8244	16.9	.2	82.9	.982
Waist girth	−.3277	.5020	.7302	21.0	32.2	46.8	.927	−.3758	.6362	.6194	23.0	39.0	38.0	.916
Neck-base girth	.2991	.4309	.1902	32.5	46.8	20.7	.894	.2989	.3381	.2833	32.5	36.7	30.8	.893
Armscye girth	.2418	.3679	.3650	24.8	37.7	37.5	.949	.1755	.3905	.4010	18.1	40.4	41.5	.944
Upper-arm girth	−.4052	.4028	.9072	23.6	23.5	52.9	.940	−.4293	.4021	.9169	24.5	23.0	52.5	.935
Posterior-arm length	.8856	.1006	−.0027	89.6	10.1	.3	.974	.9000	.0903	−.0064	90.3	9.1	.6	.975
Thigh girth	−.2585	.0875	1.1147	17.7	6.0	76.3	.970	−.2458	.1382	1.0571	17.1	9.6	73.3	.973
Maximum calf girth	.0636	.1783	.7206	6.6	18.5	74.9	.950	.0316	.1694	.7545	3.3	17.7	79.0	.946
Knee girth	.2398	.0800	.6589	24.5	8.2	67.3	.958	.1823	.1031	.6867	18.8	10.6	70.6	.954
Crotch length	.3281	.0486	.5870	34.0	5.0	61.0	.941	.2582	.0442	.6621	26.8	4.6	68.6	.944
Vertical trunk girth	.4942	.1782	.3283	49.4	17.8	32.8	.973	.4842	.1121	.4084	48.2	11.2	40.6	.977

Height and Hip Measure Recommended as Basis for Sizes

Once the fact has been established that a length and a girth (or weight) are needed as a basis for sizing garments, the problem of which length to choose is not hard to solve from both a practical and statistical point of view. A study of the multiple correlations shows that of all the possible combinations of girths (or weight) with lengths, the best combinations are waist height (from the floor) and a girth (or weight); or stature and a girth (or weight). Since waist height has a smaller range of measurement than stature and since it is more difficult to take on a child, it would seem preferable to use stature.

When a child's height is known, his other lengths can be predicted fairly accurately. For example, if a boy's stature is known, his hip height, waist height, arm length, and other lengths can be predicted with a variation of less than 1 inch on an average. The same holds true for girls. This, of course, is not surprising since some of these lengths form major parts of height.

However, the problem of which measurement to choose in order to predict the girths is slightly more complicated. The statistical analysis shows that the best predictor of all girths and lengths is either a combination of height and weight, or height and hip circumference. A poorer, but possible, combination is height and chest girth.

TABLE 7.—*Multiple correlations and percentage reduction variations of the measurements to be predicted, obtained on the basis of (1) stature and weight combination, (2) stature and hip girth combination, (3) stature and chest girth combination*

Measurements to be predicted	Boys						Girls					
	Stature—weight		Stature—hip girth		Stature—chest girth		Stature—weight		Stature—hip girth		Stature—chest girth	
	Multiple correlation	Percentage reduction	Multiple correlation	Percentage reduction	Multiple correlation	Percentage reduction	Multiple correlation	Percentage reduction	Multiple correlation	Percentage reduction	Multiple correlation	Percentage reduction
Waist height	0.990	98.1	0.990	98.1	0.991	98.1	0.990	98.0	0.990	98.1	0.990	98.1
Hip height	.985	97.0	.985	97.0	.985	97.0	.982	96.5	.982	96.5	.982	96.4
Weight			.980	96.1	.971	94.2			.985	97.0	.972	94.5
Stature												
Cervicale height	.996	99.2	.996	99.2	.996	99.3	.996	99.2	.996	99.2	.996	99.2
Tibiale height	.972	94.5	.972	94.5	.972	94.5	.968	93.6	.968	93.7	.968	93.7
Bitrochanteric diameter	.964	92.9	.977	95.4	.950	90.2	.970	94.0	.982	96.4	.955	91.3
Chest girth	.958	91.8	.951	90.4			.961	92.4	.954	91.0		
Waist girth	.915	83.7	.914	83.5	.901	81.2	.905	82.0	.896	80.2	.900	81.0
Hip girth	.974	94.9			.955	91.2	.983	96.6			.962	92.5
Neck-base girth	.893	79.7	.884	78.2	.893	79.7	.894	79.9	.887	78.7	.890	79.1
Armscye girth	.946	89.4	.942	88.7	.943	88.8	.940	88.4	.936	87.7	.937	87.8
Upper-arm girth	.930	86.5	.932	86.8	.901	81.1	.931	86.6	.927	86.0	.901	81.2
Posterior arm length	.974	94.8	.974	94.8	.974	94.9	.974	94.9	.974	94.9	.975	95.0
Thigh girth	.943	88.9	.969	93.9	.911	83.1	.961	92.3	.972	94.5	.929	86.3
Maximum calf girth	.948	89.9	.948	89.8	.925	85.6	.948	89.8	.945	89.2	.923	85.2
Knee girth	.950	90.3	.958	91.7	.938	88.0	.950	90.3	.953	90.8	.935	87.4
Crotch length	.938	88.0	.941	88.5	.925	85.5	.941	88.5	.944	89.1	.927	85.9
Vertical trunk girth	.974	94.9	.971	94.3	.968	93.6	.977	95.4	.976	95.3	.970	94.1

Table 7 gives the multiple correlations and the percentage reduction in the variations of the measurements to be predicted from the combinations of (1) stature and weight, (2) stature and girth of hips, and (3) stature and chest. The larger the multiple correlation the greater will be the percentage reduction in the variations of the measurements to be predicted. By comparing column 2 with column 4 and column 3 with 5, it will be seen that the multiple correlations and percentage reductions obtained on the combination of stature and weight of the boys do not on the whole differ from those obtained on the combination of stature and girth of hips. Thus, for the purpose of predicting the other measurements, stature and weight in combination is as effective as stature and girth of hips. On the other hand, by comparing column 4 with column

6. and column 5 with column 7. it will be noticed that the multiple correlations and percentage reduction obtained on the combination of stature and girth of hips are on the whole larger than those obtained on the combination of stature and girth of chest. Similar relationships will be observed between comparable columns relating to the girls' measurements.

Weight has been rejected for the very practical reason that every family does not possess a scale but usually has a tape measure. Hip girth has been chosen in preference to chest girth not only because of the results shown in table 7 but also for the following reasons:

A study on several thousand duplicate measurements taken by the measurers in the field shows that hip girth can be measured much more accurately than chest girth.

A study based on partial correlations (stature being held constant) has also shown that girth of hips explains the variations of the other girths better than does chest girth. This may be because girth of hips is more closely related to the fleshy and fatty parts of the body than chest girth.

The possibility of sizing garments on the basis of 3 measurements was also considered in this study even though it was realized that such a scheme would be very complicated. An analysis was undertaken to determine whether including a third measurement in the combination of 2 would yield sufficient information to warrant the sizing of garments on the bais of three measurements. This was done by setting up regression equations and calculating multiple correlations of every one of the 20 items on combination of 3 of these items.

Third dimension unnecessary

This analysis showed that introducing a third measurement into a combination of two does not increase the multiple correlation to any appreciable extent. For instance, columns 8 and 15 of table 6 give the multiple correlations on combinations of stature, chest girth, and hip girth. By comparing these columns with columns 4 and 10 of table 7, it will be noticed that the multiple correlations for corresponding measurements either remain the same when chest girth is included in the combination or are increased by a small amount. The slight increase occurs chiefly when an upper girth is predicted.

A combination of stature and hip measure is therefore recommended as a basis for sizes of patterns and garments. It should be noticed that the system suggested based on this combination refers to basic body measurements, taken next to the skin, and not to garment and patterns in which allowances must be made for undergarments.

Proposed Standard System Based on Height and Hip Measure

A system of sizes was obtained by marking off intervals of height and of hip measure on a distribution table giving the number of children having different combinations of height (stature) and hip girth. (See figs. 1 and 2 in pocket on back of cover.) The small figures on these charts represent the number of children who had a stature equal to the number of centimeters shown on the horizontal scale and a hip measure equal to the number of centimeters shown on the vertical scale.

For example, of the 835 boys who had a height of 120 centimeters, it will be found by locating the figure 120 on the horizontal scale and following it up vertically, that though they all had the same height, 1 boy had a hip measure of 50 centimeters; 1, a hip measure of 51 centimeters; 4, a hip measure of 55 centimeters; 16, a hip measure of 56 centimeters; 30, a hip measure of 57 centimeters, and so on. (Approximately 2½ centimeters is equal to 1 inch; more exactly, 1 centimeter equals 0.3937 inch.) Similarly, the distribution of the heights of the 1,889 boys who had a hip measure of 60 centimeters can be obtained by locating the figure 60 on the vertical scale and following the line across. It will be found that although all these boys measured the same around their hips, 1 had a height of 101 centimeters; 2, a height of 103 centimeters; 1, a height of 104 centimeters; and 6, a height of 106 centimeters, and so forth.

Fifty-two height-hip intervals for boys and 49 for girls were drawn on these charts in the form of rectangles. The size of these rectangles and their position are given in figures 3 and 4. They are drawn both to inch and to centimeter scale. With few exceptions almost every child can be located in the proper rectangle by reading his height on the horizontal scale and his hip measure on the vertical scale. The rectangle in which he is thus placed determines the size he takes. A size consists of averages for each of 36 body measurements. These averages were calculated on the basis of the total number of children in a given rectangle, i. e., a height-hip interval.

Thirty-nine of these height-hip intervals for boys and 36 for girls are proposed as a standard system of sizing (tables 8 and 9). These represent 89.1 percent and 86.0 percent of the samples of boys and girls, respectively. Since the remaining intervals apply to such a small percentage of children, it would not be commercially practicable to include them in a standard, and tables for them are not given here. However, they will be of assistance to those manufacturers who have a demand for garments for children whose circumferences in relation to their height are much larger or much smaller than those of the majority of children.

Regulars and auxiliary sizes

The rectangles were so chosen that the middle one for a given height interval contains the greatest number of cases. These central rectangles constitute the basis for the regular or group "C" sizes in the proposed standard. The rectangles below the middle ones constitute the basis for "below regular" or "B" sizes since the children represented in these rectangles though located in the same height interval have smaller hip circumferences than those in the middle rectangles. The rectangles lying above the middle ones constitute the basis for the "above regular" or "D" sizes. For the same height interval, the children in these rectangles have bigger hips than those in the regular rectangles.

The regular sizes suggested include 49.8 percent of the total number of boys measured and 41.8 percent of the total number of girls. The "below regular" sizes include 17.8 percent and 22.5 percent of the boys and girls, respectively, and the "above regular" sizes 21.5 percent and 21.7 percent, respectively.

Figures 3 and 4 not only give the size of the intervals but also the approximate height (in inches) and the approximate hip measure (in inches) for each interval. They also show the percentage of children in each rectangle, calculated on the basis of the total number of children in that height interval. The measurements are given in a fraction of an inch. These may be rounded off by the manufacturers when the factor of allowance is taken into consideration.

It will be noticed that for a given height interval the change in the length measurements from one size to another is very slight. This is as expected since it has been shown that, given a child's height, his lengths are fairly well determined. A change in hip measure does not substantially affect the lengths.

Some of the lengths, notably height of hips and height of waist, instead of increasing as one proceeds from one size to another, in the same height interval, tend slightly to decrease. This only bears out the analysis which shows that for a constant height the correlation of hip height or waist height with hip girth is small and negative.

TABLE 8.—*Proposed system of boys' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 3) and designated by approximate height and hip dimensions in inches*

Item No. and measurement	B_1 Height, 40½; hip, 20	C_1 Height, 40½; hip, 21½	D_1 Height, 40½; hip, 23
1. Waist height—inches	23 7/8	24	24 1/8
2. Hip height—do	19	19 1/8	19
3. Weight—pounds	33 1/2	36 1/2	39 1/2
4. Stature—inches	40 3/8	40 1/2	40 5/8
5. Cervicale height—do	32 7/8	33 1/4	33 3/8
6. Tihiale height—do	10 3/8	10 1/2	10 1/2
7. Crotch height—do	17	16 7/8	16 7/8
8. Bitrochanteric diameter—do	7 1/4	7 5/8	7 7/8
9. Shoulder slope—degrees	25 1/2	26	25 1/2
10. Anterior chest width—inches	7 1/2	7 3/4	8
11. Anterior waist length—do	8 5/8	8 5/8	8 3/4
12. Posterior chest width—do	9	9 1/4	9 1/4
13. Posterior waist length—do	9 3/8	9 3/8	9 1/2
14. Chest girth at armscye—do	21 1/4	21 7/8	22 1/2
15. Scye depth—do	4 1/2	4 5/8	4 5/8
16. Posterior hip arc—do	9 7/8	10 1/2	11 1/8
17. Maximum chest girth—do	11	11 3/8	11 3/4
18. Anterior chest arc—do	19 1/8	19 7/8	20 7/8
19. Waist girth—do	20 1/4	21 1/2	22 3/4
20. Hip girth—do	10 5/8	10 7/8	11
21. Neck-base girth—do	2 5/8	2 3/4	2 3/4
22. Shoulder length—do	9 5/8	9 7/8	10 1/4
23. Armscye girth—do	6	6 3/8	6 3/4
24. Upper-arm girth—do	6 3/4	7	7 1/4
25. Elbow girth—do	7 7/8	8	8
26. Upper posterior arm length—do	13 7/8	14 1/8	14 1/4
27. Total posterior arm length—do	4 7/8	4 7/8	4 7/8
28. Trunk line—do	5	5 1/8	5 1/8
29. Waist to hips—do	11 1/4	12 1/8	13 1/8
30. Thigh girth—do	7 7/8	8 3/8	8 3/4
31. Maximum calf girth—do	8 1/2	8 7/8	9 1/4
32. Knee girth—do	16 3/4	17 1/2	18
33. Total crotch length—do	8 1/2	8 3/4	9
34. Anterior crotch length—do	16 1/2	17	17 1/4
35. Extreme bend—do	36 3/8	37 3/8	38
36. Vertical trunk girth—do	17	61	21
37. Percentage [1]—percent			

Item No. and measurement	B_3 Height, 45; hip, 22	C_3 Height, 45; hip, 23	D_3 Height, 45½; hip, 24½
1. Waist height—inches	28	27 3/8	27 1/2
2. Hip height—do	21 7/8	21 7/8	21 7/8
3. Weight—pounds	41	44 1/2	48
4. Stature—inches	44 7/8	45 1/8	45 3/8
5. Cervicale height—do	37 3/4	37 3/8	37 1/2
6. Tihiale height—do	12	12	12
7. Crotch height—do	19 5/8	19 5/8	19 1/2
8. Bitrochanteric diameter—do	7 7/8	8 1/8	8 1/2
9. Shoulder slope—degrees	25 1/2	26	26
10. Anterior chest width—inches	8 1/8	8 1/4	8 1/2
11. Anterior waist length—do	9 1/4	9 3/8	9 1/2
12. Posterior chest width—do	9 3/4	9 7/8	10 1/8
13. Posterior waist length—do	10 1/8	10 1/4	10 3/8
14. Chest girth at armscye—do	22 1/2	23 1/4	23 7/8
15. Scye depth—do	4 7/8	5	5 1/8
16. Posterior hip arc—do	10 5/8	11 1/4	12
17. Maximum chest girth—do	11 5/8	12 1/8	12 1/2
18. Anterior chest arc—do	20	20 3/4	21 1/2
19. Waist girth—do	21 3/4	23 1/8	24 3/8
20. Hip girth—do	11 1/8	11 3/8	11 5/8
21. Neck-base girth—do	2 7/8	3	3
22. Shoulder length—do	10 1/4	10 5/8	11
23. Armscye girth—do	6 1/4	6 5/8	7
24. Upper-arm girth—do	7 1/4	7 1/2	7 3/4
25. Elbow girth—do	8 7/8	8 7/8	9
26. Upper posterior arm length—do	15 3/4	15 7/8	16
27. Total posterior arm length—do	5 3/8	5 3/8	5 3/8
28. Trunk line—do	5 1/2	5 5/8	5 3/4
29. Waist to hips—do	12 1/8	13	13 7/8
30. Thigh girth—do	8 5/8	9	9 3/8
31. Maximum calf girth—do	9 1/4	9 5/8	9 7/8
32. Knee girth—do	18 1/4	18 3/4	19 1/2
33. Total crotch length—do	9 1/8	9 1/2	9 3/4
34. Anterior crotch length—do	18 3/8	18 3/4	19 1/8
35. Extreme bend—do	39 1/4	40 1/4	41
36. Vertical trunk girth—do	14	60	23
37. Percentage [1]—percent			

Item No. and measurement	B_2 Height, 42½; hip, 21	C_2 Height, 43; hip, 22	D_2 Height, 43; hip, 23½
1. Waist height—inches	25 5/8	25 3/4	25 7/8
2. Hip height—do	20 1/2	20 1/2	20 5/8
3. Weight—pounds	37	40	44
4. Stature—inches	42 5/8	42 7/8	43 1/8
5. Cervicale height—do	35	35 1/4	35 1/2
6. Tibiale height—do	11 1/8	11 1/4	11 1/4
7. Crotch height—do	18 3/8	18 1/4	18 1/4
8. Bitrochanteric diameter—do	7 1/2	7 7/8	8 1/4
9. Shoulder slope—degrees	24 1/2	25 1/2	26
10. Anterior chest width—inches	7 3/4	8	8 1/4
11. Anterior waist length—do	8 7/8	9	9 1/8
12. Posterior chest width—do	9 1/4	9 1/4	9 3/4
13. Posterior waist length—do	9 5/8	9 7/8	10
14. Chest girth at armscye—do	21 7/8	22 1/2	23 1/4
15. Scye depth—do	4 3/4	4 3/4	4 7/8
16. Posterior hip arc—do	10 1/4	10 7/8	11 5/8
17. Maximum chest girth—do	11 3/8	11 3/4	12 1/8
18. Anterior chest arc—do	19 1/2	20 3/8	21 1/4
19. Waist girth—do	21	22 1/4	23 5/8
20. Hip girth—do	10 7/8	11 1/8	11 3/8
21. Neck-base girth—do	2 3/4	2 3/4	2 7/8
22. Shoulder length—do	9 7/8	10 1/4	10 5/8
23. Armscye girth—do	6 1/2	6 1/2	6 7/8
24. Upper-arm girth—do	7	7 1/4	7 1/2
25. Elbow girth—do	8 3/8	8 5/8	8 7/8
26. Upper posterior arm length—do	14 7/8	15	15 1/8
27. Total posterior arm length—do	5 1/8	5 1/8	5 1/8
28. Trunk line—do	5 1/4	5 3/8	5 3/8
29. Waist to hips—do	11 3/4	12 5/8	13 1/2
30. Thigh girth—do	8 1/4	8 5/8	9 1/8
31. Maximum calf girth—do	8 7/8	9 1/4	9 5/8
32. Knee girth—do	17 5/8	18	18 5/8
33. Total crotch length—do	8 7/8	9 1/4	9 1/4
34. Anterior crotch length—do	17 1/2	17 3/4	18 1/8
35. Extreme bend—do	37 7/8	38 3/4	39 1/2
36. Vertical trunk girth—do	17	60	21
37. Percentage [1]—percent			

Item No. and measurement	B_4 Height, 47; hip, 22½	C_4 Height, 47½; hip, 24	D_4 Height, 48; hip, 25
1. Waist height—inches	28 3/4	29	29 1/4
2. Hip height—do	23 1/8	23 3/8	23 3/8
3. Weight—pounds	45	49	53
4. Stature—inches	47 1/8	47 1/2	47 3/4
5. Cervicale height—do	39	39 3/8	39 5/8
6. Tihiale height—do	12 5/8	12 3/4	12 7/8
7. Crotch height—do	20 7/8	21	20 7/8
8. Bitrochanteric diameter—do	8 1/8	8 1/2	8 7/8
9. Shoulder slope—degrees	25 1/2	25 1/2	25 1/2
10. Anterior chest width—inches	8 3/8	8 5/8	8 7/8
11. Anterior waist length—do	9 5/8	9 3/4	9 7/8
12. Posterior chest width—do	10	10 1/4	10 3/8
13. Posterior waist length—do	10 5/8	10 5/8	10 3/4
14. Chest girth at armscye—do	23 1/8	24	24 5/8
15. Scye depth—do	5 1/8	5 1/4	5 1/4
16. Posterior hip arc—do	11	11 5/8	12 3/8
17. Maximum chest girth—do	12	12 1/2	13
18. Anterior chest arc—do	20 3/8	21 7/8	21 7/8
19. Waist girth—do	22 5/8	23 7/8	25 1/8
20. Hip girth—do	11 3/8	11 5/8	11 7/8
21. Neck-base girth—do	3 1/8	3 1/8	3 1/8
22. Shoulder length—do	10 5/8	11	11 3/8
23. Armscye girth—do	6 3/8	6 3/4	7 1/4
24. Upper-arm girth—do	7 1/2	7 3/4	8
25. Elbow girth—do	9 3/8	9 1/2	9 1/2
26. Upper posterior arm length—do	16 1/2	16 3/4	16 7/8
27. Total posterior arm length—do	5 5/8	5 5/8	5 5/8
28. Trunk line—do	5 3/4	5 7/8	5 7/8
29. Waist to hips—do	12 5/8	13 1/2	14 3/8
30. Thigh girth—do	8 1/2	9 1/4	9 3/4
31. Maximum calf girth—do	9 1/2	9 7/8	10 1/4
32. Knee girth—do	18 7/8	19 1/2	20 1/4
33. Total crotch length—do	9 5/8	9 7/8	10 1/8
34. Anterior crotch length—do	19 1/4	19 5/8	20
35. Extreme bend—do	40 3/4	41 5/8	42 1/2
36. Vertical trunk girth—do	12	57	27
37. Percentage [1]—percent			

See footnotes at end of table.

TABLE 8.—*Proposed system of boys' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 3) and designated by approximate height and hip dimensions in inches*—Continued

Item No. and measurement	B_5	C_5	D_5
	Height, 49½; hip, 24	Height, 50; hip, 25	Height, 50; hip, 26½
1. Waist height—inches	30½	30¾	30⅞
2. Hip height—do	24⅝	24¾	24⅞
3. Weight—pounds	50½	55	60
4. Stature—inches	49⅝	49⅞	50
5. Cervicale height—do	41⅛	41½	41¾
6. Tibiale height—do	13⅜	13½	13⅝
7. Crotch height—do	22¼	22¼	22¼
8. Bitrochanteric diameter—do	8½	8⅞	9¼
9. Shoulder slope—degrees	25½	25½	25½
10. Anterior chest width—inches	8¾	9	9¼
11. Anterior waist length—do	10	10⅛	10¼
12. Posterior chest width—do	10⅜	10⅝	10⅞
13. Posterior waist length—do	11	11⅛	11¼
14. Chest girth at armscye—do	24⅛	24⅞	25¾
15. Scye depth—do	5⅜	5⅜	5½
16. Posterior hip arc—do	11½	12⅛	13
17. Maximum chest girth—do	24¼	25⅛	26
18. Anterior chest arc—do	12⅝	13	13½
19. Waist girth—do	21	21¾	22½
20. Hip girth—do	23¾	25	26⅜
21. Neck-base girth—do	11¾	12	12¼
22. Shoulder length—do	3¼	3¼	3⅜
23. Armscye girth—do	11⅛	11½	11⅞
24. Upper-arm girth—do	6⅝	7	7½
25. Elbow girth—do	7¾	8	8⅜
26. Upper posterior arm length—do	9⅞	10	10⅛
27. Total posterior arm length—do	17½	17⅝	17⅞
28. Trunk line—do	5¾	5⅞	5⅞
29. Waist to hips—do	6	6⅛	6¼
30. Thigh girth—do	13¼	14⅛	15⅛
31. Maximum calf girth—do	9¼	9¾	10⅛
32. Knee girth—do	10	10⅜	10¾
33. Total crotch length—do	19¾	20⅜	21
34. Anterior crotch length—do	10	10¼	10⅝
35. Extreme bend—do	20⅜	20¾	21⅛
36. Vertical trunk girth—do	42⅜	43¼	44¼
37. Percentage [1]—percent	17	54	23

Item No. and measurement	B_7	C_7	D_7
	Height, 54; hip, 25½	Height, 54½; hip, 27	Height, 55; hip, 29
1. Waist height—inches	34	34¼	34½
2. Hip height—do	27⅝	27⅞	27⅞
3. Weight—pounds	62	68	75
4. Stature—inches	54¼	54½	54¾
5. Cervicale height—do	45½	45¾	46⅛
6. Tibiale height—do	15	15	15⅛
7. Crotch height—do	25⅛	25⅛	25
8. Bitrochanteric diameter—do	9¼	9⅝	10⅛
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	9½	9¾	10⅛
11. Anterior waist length—do	10⅝	10⅞	11
12. Posterior chest width—do	11	11⅜	11⅝
13. Posterior waist length—do	11¾	12	12
14. Chest girth at armscye—do	25⅞	26⅞	27⅞
15. Scye depth—do	5¾	5⅞	6
16. Posterior hip arc—do	12⅜	13¼	14¼
17. Maximum chest girth—do	26	27	28
18. Anterior chest arc—do	13½	14⅛	14¾
19. Waist girth—do	22	22⅞	23⅞
20. Hip girth—do	25⅝	27¼	28⅞
21. Neck-base girth—do	12⅜	12⅝	12⅞
22. Shoulder length—do	3½	3⅝	3⅝
23. Armscye girth—do	12⅛	12⅝	13
24. Upper-arm girth—do	7⅛	7⅝	8¼
25. Elbow girth—do	8⅜	8¾	9
26. Upper posterior arm length—do	10⅞	11⅛	11¼
27. Total posterior arm length—do	19⅜	19⅝	19¼
28. Trunk line—do	6¼	6¼	6¾
29. Waist to hips—do	6½	6½	6⅝
30. Thigh girth—do	14⅜	15½	16¾
31. Maximum calf girth—do	10	10½	11⅛
32. Knee girth—do	10⅞	11¼	11¾
33. Total crotch length—do	21¼	22	22⅞
34. Anterior crotch length—do	10¾	11⅛	11½
35. Extreme bend—do	22⅜	22⅞	23¼
36. Vertical trunk girth—do	45⅜	46⅝	47¾
37. Percentage [1]—percent	14	57	23

Item No. and measurement	B_6	C_6	D_6
	Height, 52; hip, 24½	Height, 52; hip, 26	Height, 52½; hip, 27½
1. Waist height—inches	32¼	32½	32¾
2. Hip height—do	26⅛	26¼	26⅜
3. Weight—pounds	55½	61	67½
4. Stature—inches	51⅞	52⅛	52⅜
5. Cervicale height—do	43¼	43⅝	43⅞
6. Tibiale height—do	14⅛	14¼	14⅜
7. Crotch height—do	23¾	23¾	23¾
8. Bitrochanteric diameter—do	8⅞	9¼	9¾
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	9⅛	9⅜	9¾
11. Anterior waist length—do	10¼	10½	10⅝
12. Posterior chest width—do	10¾	11	11¼
13. Posterior waist length—do	11⅜	11½	11⅝
14. Chest girth at armscye—do	24⅞	25⅞	26⅞
15. Scye depth—do	5½	5⅝	5¾
16. Posterior hip arc—do	11⅞	12⅝	13⅝
17. Maximum chest girth—do	25	26	27
18. Anterior chest arc—do	13	13½	14⅛
19. Waist girth—do	21½	22¼	23¼
20. Hip girth—do	24½	26	27⅝
21. Neck-base girth—do	12	12¼	12½
22. Shoulder length—do	3⅜	3½	3½
23. Armscye girth—do	11½	12	12½
24. Upper-arm girth—do	6¾	7¼	7⅞
25. Elbow girth—do	8	8⅜	8¾
26. Upper posterior arm length—do	10⅜	10½	10⅝
27. Total posterior arm length—do	18⅝	18⅝	18⅞
28. Trunk line—do	6	6⅛	6
29. Waist to hips—do	6¼	6¼	6⅜
30. Thigh girth—do	13⅝	14¾	16
31. Maximum calf girth—do	9½	10⅛	10⅝
32. Knee girth—do	10⅜	10¾	11¼
33. Total crotch length—do	20⅜	21⅛	21⅞
34. Anterior crotch length—do	10⅜	10⅝	11
35. Extreme bend—do	21⅜	21¾	22¼
36. Vertical trunk girth—do	43¾	44⅞	46
37. Percentage [1]—percent	13	60	23

Item No. and measurement	B_8	C_8	D_8
	Height, 56½; hip, 27	Height, 57; hip, 28½	Height, 57; hip, 30
1. Waist height—inches	35⅞	36	36⅛
2. Hip height—do	29¼	29⅜	29⅜
3. Weight—pounds	69	75½	84
4. Stature—inches	56⅝	56⅞	57
5. Cervicale height—do	47¾	48	48¼
6. Tibiale height—do	15¾	15⅞	15⅞
7. Crotch height—do	26⅝	26½	26⅜
8. Bitrochanteric diameter—do	9⅝	10⅛	10⅝
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	9⅞	10⅛	10½
11. Anterior waist length—do	11	11⅛	11¼
12. Posterior chest width—do	11⅜	11¾	11⅞
13. Posterior waist length—do	12¼	12⅜	12½
14. Chest girth at armscye—do	26⅞	27¾	28⅞
15. Scye depth—do	6	6⅛	6¼
16. Posterior hip arc—do	13	13¾	14¾
17. Maximum chest girth—do	26⅞	27⅞	29
18. Anterior chest arc—do	14⅛	14⅝	15⅜
19. Waist girth—do	22⅝	23½	24¾
20. Hip girth—do	26¾	28⅜	30⅛
21. Neck-base girth—do	12⅝	13	13¼
22. Shoulder length—do	3⅝	3¾	3⅞
23. Armscye girth—do	12⅝	13⅛	13⅝
24. Upper-arm girth—do	7⅜	7⅞	8½
25. Elbow girth—do	8¾	9	9⅜
26. Upper posterior arm length—do	11½	11⅝	11¾
27. Total posterior arm length—do	20⅜	20⅝	20¾
28. Trunk line—do	6½	6½	6½
29. Waist to hips—do	6¾	6⅞	6⅞
30. Thigh girth—do	15	16⅛	17⅝
31. Maximum calf girth—do	10⅜	11	11⅝
32. Knee girth—do	11¼	11¾	12¼
33. Total crotch length—do	22¼	22⅞	23¾
34. Anterior crotch length—do	11¼	11⅝	12
35. Extreme bend—do	23⅝	23⅞	24⅜
36. Vertical trunk girth—do	47¼	48⅜	49⅝
37. Percentage [1]—percent	15	52	24

See footnotes at end of table.

TABLE 8.—*Proposed system of boys' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 3) and designated by approximate height and hip dimensions in inches*—Continued

Item No. and measurement	B_9 Height, 59; hip, 28	C_9 Height, 59; hip, 30	D_9 Height, 59½; hip, 32
1. Waist height—inches	37⅝	37¾	37⅞
2. Hip height—do	30⅜	30⅞	30⅞
3. Weight—pounds	78	86	95½
4. Stature—inches	59	59¼	59⅜
5. Cervicale height—do	50	50¼	50⅜
6. Tibiale height—do	16⅝	16⅝	16¾
7. Croteh height—do	27⅞	27⅞	27⅝
8. Bitrochanteric diameter—do	10⅛	10⅜	11⅛
9. Shoulder slope—degrees	24½	24½	25
10. Anterior chest width—inches	10⅜	10⅝	11
11. Anterior waist length—do	11⅜	11½	11⅝
12. Posterior ehest widtb—do	11¾	12⅛	12⅜
13. Posterior waist length—do	12⅜	12⅞	13
14. Chest girth at armseye—do	28	29⅛	30⅜
15. Scye depth—do	6¼	6⅜	6½
16. Posterior hip are—do	13⅝	14½	15½
17. Maximum chest girth—do	28	29⅛	30⅜
18. Anterior chest are—do	14¾	15⅜	16⅛
19. Waist girth—do	23½	24⅜	25¾
20. Hip girth—do	28¼	29⅞	31¾
21. Neck-hase girth—do	13	13⅜	13⅝
22. Shoulder length—do	3⅞	3⅞	4
23. Armseye girth—do	13¼	13¾	14⅜
24. Upper-arm girth—do	7¾	8⅜	9
25. Elbow girth—do	9⅛	9½	9⅞
26. Upper posterior arm length—do	12⅛	12⅝	12¾
27. Total posterior arm length—do	21⅜	21⅝	21¾
28. Trunk line—do	6¾	6¾	6⅝
29. Waist to hips—do	7	7⅛	7½
30. Thigh girth—do	15⅞	17⅛	18⅝
31. Maximum ealf girth—do	11	11½	12¼
32. Knee girth—do	11⅞	12⅜	12⅞
33. Total crotch length—do	23¼	24⅛	25⅛
34. Anterior crotch length—do	11⅞	12¼	12¾
35. Extreme bend—do	24⅝	25	25½
36. Vertical trunk girth—do	49¼	50⅝	52
37. Percentage [1]—percent	23	47	21

Item No. and measurement	B_{11} Height, 64; hip, 31½	C_{11} Height, 64; bip, 33	D_{11} Height, 64; hip, 35
1. Waist height—ineches	40¾	40¾	40⅝
2. Hip height—do	33½	33⅜	33⅛
3. Weight—pounds	100½	111	123
4. Stature—inches	63¾	64	64¼
5. Cervicale height—do	54½	54⅝	54⅞
6. Tibiale height—do	18⅛	18	17⅞
7. Crotch height—do	30⅜	30	29¾
8. Bitrochanterie diameter—do	11¼	11⅞	12⅜
9. Shoulder slope—degrees	24	25	25½
10. Anterior ehest width—ineches	11½	11⅞	12⅜
11. Anterior waist length—do	12⅜	12⅝	12⅞
12. Posterior chest width—do	12¾	13¼	13⅝
13. Posterior waist length—do	14	14¼	14⅝
14. Cbest girth at armseye—do	30¾	32⅜	33⅞
15. Scye depth—do	6⅞	7	7¼
16. Posterior hip arc—do	15⅛	16	17
17. Maximum chest girth—do	30¾	32¼	33⅞
18. Anterior chest arc—do	16¼	17	17⅞
19. Waist girth—do	25⅛	26¼	27⅝
20. Hip girth—do	31⅜	33	34⅞
21. Neck-base girth—do	14⅛	14⅝	15⅛
22. Shoulder length—do	4¼	4⅜	4½
23. Armseye girth—do	14¾	15⅜	16
24. Upper-arm girth—do	8¾	9⅜	10⅛
25. Elbow girth—do	10⅛	10½	11
26. Upper posterior arm length—do	13⅛	13¼	13⅜
27. Total posterior arm length—do	23⅜	23½	23⅝
28. Trunk line—do	7⅜	7½	7½
29. Waist to hips—do	7½	7⅝	7¾
30. Thigh girth—do	17⅝	18⅞	20¼
31. Maximum calf girth—do	12⅛	12⅜	13½
32. Knee girtb—do	13	13⅜	13⅞
33. Total croteh length—do	25⅞	26⅜	27⅝
34. Anterior eroteh length—do	13¼	13⅜	14⅛
35. Extreme bend—do	26⅞	27⅛	27½
36. Vertical trunk girth—do	54⅜	56	57⅝
37. Percentage [1]—pereent	25	42	22

Item No. and measurement	B_{10} Height, 61½; hip, 30	C_{10} Height, 61½; hip, 31½	D_{10} Height, 62; hip, 33
1. Waist height—inebes	39⅜	39⅜	39⅜
2. Hip height—do	32¼	32¼	32
3. Weight—pounds	88	98	108
4. Stature—inches	61⅜	61⅝	61¾
5. Cervicale height—do	52¼	52½	52⅝
6. Tibiale height—do	17⅜	17⅜	17⅜
7. Crotcb height—do	29¼	29	28¾
8. Bitrochanterie diameter—do	10¾	11¼	11¾
9. Shoulder slope—degrees	24½	25	25
10. Anterior chest width—inches	10⅞	11¼	11⅝
11. Anterior waist length—do	11¾	12	12⅛
12. Posterior ehest width—do	12¼	12⅝	12⅞
13. Posterior waist length—do	13¼	13½	13¾
14. Chest girth at armseye—do	29⅜	30½	31⅞
15. Scye depth—do	6½	6⅝	6¾
16. Posterior hip are—do	14⅜	15¼	16¼
17. Maximum ehest girth—do	29¼	30½	31⅞
18. Anterior ehest are—do	15½	16¼	17
19. Waist girth—do	24⅜	25⅜	26⅝
20. Hip girth—do	29¾	31½	33¼
21. Neck-base girth—do	13½	13⅞	14¼
22. Shoulder length—do	4	4⅛	4⅛
23. Armseye girth—do	14	14½	15⅛
24. Upper-arm girtb—do	8¼	8⅞	9½
25. Elbow girth—do	9⅜	10	10⅜
26. Upper posterior arm length—do	12⅝	12¾	12¾
27. Total posterior arm length—do	22⅜	22½	22¾
28. Trunk line—do	7	7	7
29. Waist to hips—do	7¼	7⅜	7½
30. Thigh girth—do	16⅞	18	19½
31. Maximum calf girth—do	11½	12¼	12⅞
32. Knee girth—do	12½	13	13½
33. Total crotch length—do	24⅜	25⅜	26⅜
34. Anterior crotch length—do	12½	13	13⅜
35. Extreme bend—do	25¾	26½	26½
36. Vertical trunk girth—do	51½	53⅛	54⅝
37. Pereentage [1]—percent	26	43	20

Item No. and measurement	B_{12} Height, 66; hip, 33	C_{12} Height, 66½; hip, 34½	D_{12} Height, 66½; bip, 36½
1. Waist height—inches	42⅛	42	41⅞
2. Hip beight—do	34⅝	34⅜	34¼
3. Weight—pounds	113	125½	138
4. Stature—inches	66¼	66⅜	66½
5. Cervicale heighet—do	56⅜	56¾	56⅞
6. Tibiale beight—do	18⅝	18½	18½
7. Croteh height—do	31¼	30⅞	30⅝
8. Bitrochanterie diameter—do	12	12⅜	12⅞
9. Shoulder slope—degrees	25	25	25
10. Anterior ehest width—ineches	12⅛	12⅝	13
11. Anterior waist length—do	12⅞	13⅛	13⅜
12. Posterior chest width—do	13⅜	13⅞	14⅜
13. Posterior waist length—do	14¾	15⅛	15⅜
14. Cbest girth at armseye—do	32½	34⅛	35⅝
15. Scye depth—do	7¼	7⅜	7½
16. Posterior hip arc—do	15⅞	16⅞	17⅞
17. Maximum ehest girth—do	32½	34⅛	35¾
18. Anterior ehest arc—do	17⅛	18	18⅞
19. Waist girth—do	26⅛	27¼	28⅝
20. Hip girth—do	32⅞	34⅝	36½
21. Neck-base girth—do	14⅞	15⅜	15⅞
22. Shoulder length—do	4½	4½	4⅝
23. Armseye girtb—do	15½	16⅛	16¾
24. Upper-arm girth—do	9¼	10	10¾
25. Elbow girth—do	10½	11	11⅜
26. Upper posterior arm length—do	13¾	13⅞	13⅞
27. Total posterior arm length—do	24¼	24⅜	24⅝
28. Trunk line—do	7¾	7⅞	7⅞
29. Waist to hips—do	7⅞	8	8
30. Thigb girth—do	18½	19¾	21⅛
31. Maximum ealf girth—do	12¾	13⅜	14
32. Knee girth—do	13⅜	13¾	14¼
33. Total erotch length—do	27⅛	27⅞	28¾
34. Anterior crotch length—do	14	14⅜	14⅝
35. Extreme bend—do	27⅞	28¼	28⅝
36. Vertical trunk girth—do	57⅛	58¾	60⅛
37. Pereentage [1]—pereent	25	44	20

See footnotes at end of table.

TABLE 8.—*Proposed system of boys' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 3) and designated by approximate height and hip dimensions in inches*—Continued

Item No. and measurement	B_{13}	C_{13}	D_{13}
	Height, 68½; hip, 34	Height, 68½; hip, 36	Height, 69; hip, 37½
1. Waist height—inches	43½	43⅜	43⅜
2. Hip height—do	35¾	35½	35⅜
3. Weight—pounds	124½	136½	151
4. Stature—inches	68½	68⅝	68¾
5. Cervicale height—do	58⅝	58¾	58⅞
6. Tibiale height—do	19⅛	19⅛	19⅛
7. Crotch height—do	32¼	31⅞	31¾
8. Bitrochanteric diameter—do	12⅜	12⅞	13¼
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	12⅝	13	13⅜
11. Anterior waist length—do	13½	13⅜	13⅞
12. Posterior chest width—do	13⅞	14¼	14¾
13. Posterior waist length—do	15⅜	15¾	15⅞
14. Chest girth at armscye—do	33¾	35¼	36¾
15. Scye depth—do	7⅝	7¾	7⅞
16. Posterior hip arc—do	16½	17½	18½
17. Maximum chest girth—do	33¾	35¼	36¾
18. Anterior chest arc—do	17¾	18½	19⅜
19. Waist girth—do	26⅝	27⅞	29⅜
20. Hip girth—do	34	35¾	37⅝
21. Neck-base girth—do	15⅜	15⅞	16⅜
22. Shoulder length—do	4⅝	4¾	4¾
23. Armscye girth—do	16⅛	16¾	17⅜
24. Upper-arm girth—do	9⅝	10⅜	11
25. Elbow girth—do	10⅞	11⅜	11⅝
26. Upper posterior arm length—do	14¼	14¼	14⅜
27. Total posterior arm length—do	25⅛	25¼	25⅜
28. Trunk line—do	8⅛	8⅛	8⅛
29. Waist to hips—do	8⅛	8¼	8¼
30. Thigh girth—do	19	20⅜	21¾
31. Maximum calf girth—do	13⅛	13¾	14⅜
32. Knee girth—do	13⅝	14⅛	14⅝
33. Total crotch length—do	28⅛	28⅞	29⅝
34. Anterior crotch length—do	14⅜	14¾	15⅛
35. Extreme bend—do	28⅞	29⅛	29⅝
36. Vertical trunk girth—do	59⅜	60¾	62⅛
37. Percentage [1]—percent	28	44	18

[1] Based on total number of children in the given height interval.

TABLE 9.—*Proposed system of girls' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 4) and designated by approximate height and hip dimensions in inches*

Item No. and measurement	B_1	C_1	D_1
	Height, 40½; hip, 20½	Height, 40½; hip, 22	Height, 41; hip, 23½
1. Waist height—inches	24¼	24⅜	24⅝
2. Hip height—do	19¼	19⅜	19½
3. Weight—pounds	33	36	39½
4. Stature—inches	40⅜	40⅝	40¾
5. Cervicale height—do	33⅛	33⅜	33⅝
6. Tihiale height—do	10⅝	10⅝	10¾
7. Crotch height—do	17	17⅛	17
8. Bitrochanteric diameter—do	7¼	7⅝	8
9. Shoulder slope—degrees	25½	25½	26
10. Anterior chest width—inches	7⅝	7⅞	8
11. Anterior waist length—do	8½	8⅝	8⅝
12. Posterior chest width—do	8⅞	9⅛	9⅜
13. Posterior waist length—do	9¼	9¼	9⅜
14. Chest girth at armscye—do	20⅞	21⅝	22⅜
15. Scye depth—do	4⅛	4½	4⅝
16. Posterior hip arc—do	10⅛	10⅞	11⅜
17. Maximum chest girth—do	10⅞	11¼	11⅝
18. Anterior chest arc—do	18¾	19½	20⅜
19. Waist girth—do	20½	21⅞	23⅛
20. Hip girth—do	10½	10⅝	10⅞
21. Neck-base girth—do	2⅝	2¾	2⅞
22. Shoulder length—do	9⅜	9¾	10⅛
23. Armscye girth—do	6	6⅜	6¾
24. Upper-arm girth—do	6⅝	6⅞	7⅛
25. Elbow girth—do	7	7⅛	7⅜
26. Upper posterior arm length—do	13⅞	14⅛	14⅛
27. Total posterior arm length—do	4⅞	4⅞	4¾
28. Trunk line—do	5⅛	5¼	5⅜
29. Waist to hips—do	11⅝	12⅝	13½
30. Thigh girth—do	8⅛	8½	8⅞
31. Maximum calf girth—do	8½	8⅞	9¼
32. Knee girth—do	17	17½	18⅛
33. Total crotch length—do	8½	8⅝	9
34. Anterior crotch length—do	17	17⅜	17¾
35. Extreme hend—do	36⅛	36⅞	37⅝
36. Vertical trunk girth—do	25	56	17
37. Percentage [1]—pereent			

Item No. and measurement	B_3	C_3	D_3
	Height, 45; hip, 22	Height, 45; hip, 23½	Height, 45½; hip, 25
1. Waist height—inches	27½	27¾	27⅞
2. Hip height—do	22⅛	22⅛	22¼
3. Weight—pounds	40	44	48
4. Stature—inches	44⅞	45⅛	45⅜
5. Cervicale height—do	37¼	37½	37¾
6. Tihiale height—do	12	12⅛	12¼
7. Crotch height—do	19¾	19¾	19¾
8. Bitrochanteric diameter—do	7⅞	8¼	8⅜
9. Shoulder slope—degrees	25	25½	26
10. Anterior chest width—inches	8⅛	8⅜	8⅝
11. Anterior waist length—do	9⅛	9¼	9⅜
12. Posterior chest width—do	9⅝	9¾	10
13. Posterior waist length—do	10	10⅛	10¼
14. Chest girth at armscye—do	22⅛	22⅞	23⅝
15. Scye depth—do	4⅞	5	5
16. Posterior hip arc—do	10⅝	11⅝	12⅜
17. Maximum chest girth—do	11½	11⅞	12⅜
18. Anterior chest arc—do	19⅜	20⅛	21
19. Waist girth—do	22⅛	23⅜	24¾
20. Hip girth—do	10⅝	11⅛	11⅜
21. Neck-base girth—do	3	3⅛	3⅛
22. Shoulder length—do	10	10⅜	10¾
23. Armscye girth—do	6¼	6⅝	7⅛
24. Upper-arm girth—do	7	7⅛	7⅜
25. Elbow girth—do	8⅞	9	9
26. Upper posterior arm length—do	15⅝	15¾	15⅞
27. Total posterior arm length—do	5⅜	5⅝	5¼
28. Trunk line—do	5⅝	5⅝	5¾
29. Waist to hips—do	12½	13⅜	14½
30. Thigh girth—do	8⅝	9⅛	9⅝
31. Maximum calf girth—do	9⅛	9⅝	10
32. Knee girth—do	18⅜	18⅞	19½
33. Total crotch length—do	9⅛	9⅜	9⅝
34. Anterior crotch length—do	18⅞	19¼	19⅝
35. Extreme hend—do	38⅞	39⅝	40⅜
36. Vertical trunk girth—do	19	55	21
37. Percentage [1]—percent			

Item No. and measurement	B_2	C_2	D_2
	Height, 42½; hip, 21½	Height, 43; hip, 22½	Height, 43; hip, 24
1. Waist height—inches	25⅞	26⅛	26¼
2. Hip height—do	20⅝	20¾	20⅞
3. Weight—pounds	36½	40	43½
4. Stature—inches	42⅝	42⅞	43
5. Cervicale height—do	35⅛	35½	35⅝
6. Tihiale height—do	11¼	11⅜	11½
7. Crotch height—do	18⅜	18⅜	18⅜
8. Bitrochanteric diameter—do	7⅝	8	8⅜
9. Shoulder slope—degrees	25	25½	25½
10. Anterior chest width—inches	7¾	8	8¼
11. Anterior waist length—do	8⅞	8⅞	9
12. Posterior chest width—do	9⅜	9⅜	9⅝
13. Posterior waist length—do	9⅝	9⅝	9¾
14. Chest girth at armscye—do	21⅝	22⅛	22⅞
15. Scye depth—do	4⅝	4¾	4¾
16. Posterior hip arc—do	10½	11¼	11⅞
17. Maximum chest girth—do	11¼	11⅝	12
18. Anterior chest arc—do	19	19¾	20¾
19. Waist girth—do	21⅜	22⅝	24
20. Hip girth—do	10¾	10⅞	11⅛
21. Neck-base girth—do	2⅞	2⅞	3
22. Shoulder length—do	9¾	10	10⅜
23. Armscye girth—do	6⅛	6½	7
24. Upper-arm girth—do	6⅞	7⅛	7⅜
25. Elbow girth—do	8⅜	8½	8½
26. Upper posterior arm length—do	14¾	14⅞	15
27. Total posterior arm length—do	5⅛	5⅛	5
28. Trunk line—do	5⅜	5⅜	5½
29. Waist to hips—do	12⅛	13	14
30. Thigh girth—do	8⅜	8¾	9¼
31. Maximum calf girth—do	8⅞	9¼	9⅝
32. Knee girth—do	17¾	18¼	18⅞
33. Total crotch length—do	8¾	9	9¼
34. Anterior crotch length—do	17⅛	18¼	18¾
35. Extreme hend—do	37⅜	38¼	39⅛
36. Vertical trunk girth—do	23	56	18
37. Percentage [1]—percent			

Item No. and measurement	B_4	C_4	D_4
	Height, 47; hip, 23	Height, 47½; hip, 24	Height, 47½; hip, 25½
1. Waist height—inches	29⅛	29⅜	29⅝
2. Hip height—do	23⅜	23⅝	23⅞
3. Weight—pounds	44½	48	53
4. Stature—inches	47¼	47½	47⅝
5. Cervicale height—do	39¼	39½	39¾
6. Tihiale height—do	12¾	12⅞	13
7. Crotch height—do	21	21	21
8. Bitrochanteric diameter—do	8¼	8½	9
9. Shoulder slope—degrees	25	25½	26
10. Anterior chest width—inches	8⅜	8⅝	8⅞
11. Anterior waist length—do	9½	9⅝	9¾
12. Posterior chest width—do	9⅞	10⅛	10¼
13. Posterior waist length—do	10⅜	10½	10⅝
14. Chest girth at armscye—do	22⅞	23½	24⅜
15. Scye depth—do	5	5⅛	5¼
16. Posterior hip arc—do	11⅜	12	12¾
17. Maximum chest girth—do	22⅞	23½	24⅜
18. Anterior chest arc—do	11⅞	12⅜	12⅞
19. Waist girth—do	19¾	20½	21⅜
20. Hip girth—do	22⅞	24¼	25⅝
21. Neck-base girth—do	11⅛	11⅝	11⅝
22. Shoulder length—do	3⅛	3¼	3¼
23. Armscye girth—do	10⅜	10¾	11
24. Upper-arm girth—do	6⅜	6¾	7¼
25. Elbow girth—do	7¼	7½	7¾
26. Upper posterior arm length—do	9⅜	9½	9½
27. Total posterior arm length—do	16⅜	16⅝	16¾
28. Trunk line—do	5⅝	5⅝	5½
29. Waist to hips—do	5⅞	5⅞	6
30. Thigh girth—do	13	13⅞	14⅞
31. Maximum calf girth—do	9	9⅜	9⅞
32. Knee girth—do	9½	9⅞	10⅜
33. Total crotch length—do	19⅛	19⅝	20⅜
34. Anterior crotch length—do	9½	9¾	10
35. Extreme hend—do	19⅞	20¼	20⅝
36. Vertical trunk girth—do	40¼	41	41⅞
37. Percentage [1]—percent	17	50	26

See footnotes at end of table.

TABLE 9.—*Proposed system of girls' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 4) and designated by approximate height and hip dimensions in inches*—Continued

Item No. and measurement	B_5 Height, 49½; hip, 24	C_5 Height, 50; hip, 25½	D_5 Height, 50; hip, 27
1. Waist height—inches	31	31⅛	31⅜
2. Hip height—do	25	25⅛	25⅛
3. Weight—pounds	49½	54	59
4. Stature—inches	49⅝	49⅞	50
5. Cervicale height—do	41⅜	41⅝	41¾
6. Tibiale height—do	13⅝	13⅝	13¾
7. Crotch height—do	22⅜	22⅜	22⅜
8. Bitrochanteric diameter—do	8⅝	9	9⅜
9. Shoulder slope—degrees	25	25½	25½
10. Anterior chest width—inches	8¾	9	9⅜
11. Anterior waist length—do	9⅞	10	10
12. Posterior chest width—do	10¼	10½	10⅝
13. Posterior waist length—do	10⅞	10⅞	11
14. Chest girth at armscye—do	23⅝	24⅜	25¼
15. Scye depth—do	5¼	5⅜	5⅜
16. Posterior hip arc—do	11⅞	12½	13⅜
17. Maximum chest girth—do	23¾	24½	25¼
18. Anterior chest arc—do	12⅜	12⅞	13⅜
19. Waist girth—do	20¼	21⅛	22
20. Hip girth—do	24	25⅜	26⅞
21. Neck-base girth—do	11½	11¾	11⅞
22. Shoulder length—do	3⅜	3⅜	3½
23. Armscye girth—do	10¾	11⅛	11½
24. Upper-arm girth—do	6⅝	7	7½
25. Elbow girth—do	7½	7¾	8⅛
26. Upper posterior arm length—do	9⅞	10	10
27. Total posterior arm length—do	17⅜	17⅝	17¾
28. Trunk line—do	5¾	5¾	5¾
29. Waist to hips—do	6½	6¼	6⅜
30. Thigh girth—do	13⅝	14⅝	15⅝
31. Maximum calf girth—do	9⅜	9⅞	10¼
32. Knee girth—do	10	10⅜	10¾
33. Total crotch length—do	20	20⅝	21¼
34. Anterior crotch length—do	10	10¼	10½
35. Extreme bend—do	21	21⅜	21¾
36. Vertical trunk girth—do	41⅞	42⅝	43½
37. Percentage [1]—percent	19	47	24

Item No. and measurement	B_7 Height, 54½; hip, 26½	C_7 Height, 54½; hip, 28	D_7 Height, 54½; hip, 30
1. Waist height—inches	34½	34⅝	34⅞
2. Hip height—do	28	28	28¼
3. Weight—pounds	63	69	77½
4. Stature—inches	54⅜	54½	54⅝
5. Cervicale height—do	45⅞	46	46¼
6. Tibiale height—do	15⅛	15¼	15¼
7. Crotch height—do	25⅛	25⅛	25
8. Bitrochanteric diameter—do	9½	10	10½
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	9⅝	9⅞	10¼
11. Anterior waist length—do	10½	10⅝	10¾
12. Posterior chest width—do	11	11¼	11½
13. Posterior waist length—do	11¾	11⅞	11⅞
14. Chest girth at armscye—do	25⅝	26½	27⅞
15. Scye depth—do	5⅝	5¾	6
16. Posterior hip arc—do	13⅛	14	15
17. Maximum chest girth—do	25⅝	26½	27⅞
18. Anterior chest arc—do	13½	14⅛	14⅞
19. Waist girth—do	21½	22⅝	24
20. Hip girth—do	26⅝	28¼	30⅛
21. Neck-base girth—do	12¼	12⅜	12¾
22. Shoulder length—do	3⅝	3¾	3¾
23. Armscye girth—do	11¾	12¼	12¾
24. Upper-arm girth—do	7¼	7¾	8½
25. Elbow girth—do	8¼	8½	8⅞
26. Upper posterior arm length—do	11	11⅛	11¼
27. Total posterior arm length—do	19⅜	19½	19⅝
28. Trunk line—do	6¼	6¼	6½
29. Waist to hips—do	6¾	6⅞	7
30. Thigh girth—do	15½	16⅜	17¾
31. Maximum calf girth—do	10¼	10¾	11⅜
32. Knee girth—do	10⅞	11⅜	12
33. Total crotch length—do	21⅞	22⅝	23⅜
34. Anterior crotch length—do	10⅞	11⅛	11⅝
35. Extreme bend—do	23¼	23¾	24¼
36. Vertical trunk girth—do	45⅜	46½	47¾
37. Percentage [1]—percent	29	43	18

Item No. and measurement	B_6 Height, 52; hip, 25	C_6 Height, 52; hip, 27	D_6 Height, 52; hip, 28½
1. Waist height—inches	32¾	32⅞	33¼
2. Hip height—do	26⅜	26⅝	26⅝
3. Weight—pounds	55½	61	68
4. Stature—inches	51⅞	52⅛	52¼
5. Cervicale height—do	43½	43⅞	44
6. Tibiale height—do	14⅜	14½	14½
7. Crotch height—do	23¾	23¾	23¾
8. Bitrochanteric diameter—do	9	9⅜	9⅝
9. Shoulder slope—degrees	25	25	25½
10. Anterior chest width—inches	9¼	9⅜	9¾
11. Anterior waist length—do	10¼	10¼	10⅜
12. Posterior chest width—do	10⅝	10⅞	11
13. Posterior waist length—do	11¼	11⅜	11⅜
14. Chest girth at armscye—do	24½	25⅜	26½
15. Scye depth—do	5½	5½	5⅝
16. Posterior hip arc—do	12⅜	13¼	14¼
17. Maximum chest girth—do	24⅝	25⅜	26⅝
18. Anterior chest arc—do	12⅞	13½	14⅛
19. Waist girth—do	20⅞	21¾	23⅛
20. Hip girth—do	25⅞	26¾	28½
21. Neck-base girth—do	11⅞	12	12¼
22. Shoulder length—do	3½	3⅝	3⅝
23. Armscye girth—do	11¼	11⅝	12⅛
24. Upper-arm girth—do	6⅞	7⅜	8
25. Elbow girth—do	7⅞	8⅛	8½
26. Upper posterior arm length—do	10½	10½	10⅝
27. Total posterior arm length—do	18⅜	18½	18¾
28. Trunk line—do	6	6	5⅞
29. Waist to hips—do	6⅜	6½	6⅝
30. Thigh girth—do	14¼	15⅜	16¾
31. Maximum calf girth—do	9¾	10¼	10⅞
32. Knee girth—do	10⅜	11	11⅜
33. Total crotch length—do	20⅞	21⅝	22½
34. Anterior crotch length—do	10⅜	10⅝	11⅛
35. Extreme bend—do	22⅛	22½	23
36. Vertical trunk girth—do	43⅝	44½	45⅝
37. Percentage [1]—percent	21	51	21

Item No. and measurement	B_8 Height, 57; hip, 28	C_8 Height, 57; hip, 30	D_8 Height, 57; hip, 32
1. Waist height—inches	36¼	36⅜	36½
2. Hip height—do	29⅜	29⅜	29⅜
3. Weight—pounds	71	78	87½
4. Stature—inches	56¾	56⅞	57
5. Cervicale height—do	48	48¼	48½
6. Tibiale height—do	15⅞	16	16
7. Crotch height—do	26¼	26¼	26
8. Bitrochanteric diameter—do	10	10½	11⅛
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	10	10¼	10⅝
11. Anterior waist length—do	10⅞	11	11⅛
12. Posterior chest width—do	11½	11⅝	12
13. Posterior waist length—do	12¼	12⅜	12½
14. Chest girth at armscye—do	26¾	27⅞	29¼
15. Scye depth—do	5⅞	6	6⅛
16. Posterior hip arc—do	13⅞	14¾	15¾
17. Maximum chest girth—do	26⅝	27⅞	29⅜
18. Anterior chest arc—do	14⅛	14¾	15½
19. Waist girth—do	22⅜	23⅜	24¾
20. Hip girth—do	28½	29⅞	31¾
21. Neck-base girth—do	12⅝	12⅞	13⅛
22. Shoulder length—do	3⅞	3⅞	4
23. Armscye girth—do	12⅜	12⅞	13⅜
24. Upper-arm girth—do	7⅜	8⅛	8⅞
25. Elbow girth—do	8⅝	9	9¼
26. Upper posterior arm length—do	11⅝	11⅝	11¾
27. Total posterior arm length—do	20⅜	20⅝	20⅝
28. Trunk line—do	6½	6½	6⅜
29. Waist to hips—do	7	7¼	7⅜
30. Thigh girth—do	16	17¼	18⅝
31. Maximum calf girth—do	10⅜	11¼	11⅞
32. Knee girth—do	11½	12	12½
33. Total crotch length—do	22⅞	23⅜	24¾
34. Anterior crotch length—do	11⅜	11⅝	12⅛
35. Extreme bend—do	24½	24⅞	25⅜
36. Vertical trunk girth—do	47⅝	48⅝	50
37. Percentage [1]—percent	30	36	19

See footnotes at end of table.

TABLE 9.—*Proposed system of girls' body measurements based on intervals of height and hip measure (symbols, i. e., B_1, C_1, D_1, show corresponding intervals on fig. 4) and designated by approximate height and hip dimensions in inches*—Continued

Item No. and measurement	B_9	C_9	D_9
	Height, 59; hip, 30	Height, 59½; hip, 32	Height, 59½; hip, 34
1. Waist height—inches	37⅞	38	38
2. Hip height—do	30⅝	30⅝	30⅝
3. Weight—pounds	81½	90	99½
4. Stature—inches	59⅜	59⅜	59½
5. Cervicale height—do	50¼	50½	50¾
6. Tibiale height—do	16⅝	16⅝	16½
7. Crotch height—do	27½	27¼	27⅛
8. Bitrochanteric diameter—do	10¾	11¼	11⅞
9. Shoulder slope—degrees	25	25	25
10. Anterior chest width—inches	10½	10⅞	11¼
11. Anterior waist length—do	11⅜	11½	11¾
12. Posterior chest width—do	11⅞	12¼	12½
13. Posterior waist length—do	12⅞	13⅛	13⅜
14. Chest girth at armscye—do	28⅛	29⅜	30¾
15. Scye depth—do	6¼	6⅜	6⅜
16. Posterior hip arc—do	14¾	15¾	16⅞
17. Maximum chest girth—do	28¼	29⅜	31¼
18. Anterior chest arc—do	14⅞	15½	16⅜
19. Waist girth—do	23¼	24⅛	25
20. Hip girth—do	30	31⅞	33¾
21. Neck-base girth—do	13	13⅜	13¾
22. Shoulder length—do	4	4⅛	4⅛
23. Armscye girth—do	13	13½	14
24. Upper-arm girth—do	8	8½	9¼
25. Elbow girth—do	9	9⅜	9⅝
26. Upper posterior arm length—do	12⅛	12¼	12⅜
27. Total posterior arm length—do	21⅜	21½	21⅝
28. Trunk line—do	6⅞	6⅞	6⅞
29. Waist to hips—do	7⅜	7⅜	7⅞
30. Thigh girth—do	17	18¼	19⅜
31. Maximum calf girth—do	11¼	11⅞	12⅜
32. Knee girth—do	12	12½	12⅞
33. Total crotch length—do	24½	25	26
34. Anterior crotch length—do	11⅞	12⅜	12¾
35. Extreme bend—do	25⅝	26⅛	26⅝
36. Vertical trunk girth—do	49⅞	51⅛	52¾
37. Percentage [1]—percent	26	30	21

Item No. and measurement	B_{10}	C_{10}	D_{10}
	Height, 61½; hip, 32	Height, 61½; hip, 34	Height, 62; hip, 36
1. Waist height—inches	39¼	39¼	39¼
2. Hip height—do	31¾	31⅝	31½
3. Weight—pounds	92½	102	112
4. Stature—inches	61½	61⅝	61¾
5. Cervicale height—do	52½	52⅝	52¾
6. Tibiale height—do	17¼	17⅛	17
7. Crotch height—do	28⅝	28¼	28
8. Bitrochanteric diameter—do	11½	12	12⅝
9. Shoulder slope—degrees	25	25	24
10. Anterior chest width—inches	11	11⅜	11¾
11. Anterior waist length—do	11⅞	12⅛	12¼
12. Posterior chest width—do	12⅜	12⅝	12⅞
13. Posterior waist length—do	13⅝	13⅞	14
14. Chest girth at armscye—do	29½	30⅞	32⅛
15. Scye depth—do	6½	6¾	6⅞
16. Posterior hip arc—do	15¾	16¾	17¾
17. Maximum chest girth—do	29⅞	31½	32¾
18. Anterior chest arc—do	15⅝	16½	17¼
19. Waist girth—do	23⅞	24⅝	25⅝
20. Hip girth—do	32	33⅞	35¾
21. Neck-base girth—do	13½	13⅞	14⅛
22. Shoulder length—do	4⅛	4¼	4¼
23. Armscye girth—do	13⅝	14⅛	14⅝
24. Upper-arm girth—do	8⅜	9	9⅝
25. Elbow girth—do	9⅜	9⅝	9⅞
26. Upper posterior arm length—do	12¾	12⅞	12⅞
27. Total posterior arm length—do	22¼	22⅜	22½
28. Trunk line—do	7¼	7¼	7¼
29. Waist to hips—do	7⅞	8	8¼
30. Thigh girth—do	18⅛	19⅜	20¾
31. Maximum calf girth—do	11⅞	12⅜	13
32. Knee girth—do	12½	12⅞	13⅜
33. Total crotch length—do	25½	26¼	27⅛
34. Anterior crotch length—do	12½	12⅞	13⅜
35. Extreme bend—do	26⅞	27¼	27¾
36. Vertical trunk girth—do	52¼	53¾	55
37. Percentage [1]—percent	20	32	26

Item No. and measurement	B_{11}	C_{11}	D_{11}
	Height, 64; hip, 33½	Height, 64; hip, 35½	Height, 64; hip, 37
1. Waist height—inches	40⅝	40⅝	40¾
2. Hip height—do	32⅞	32¾	32¾
3. Weight—pounds	103	112½	123½
4. Stature—inches	63¾	63⅞	64
5. Cervicale height—do	54½	54⅝	54¾
6. Tibiale height—do	17¾	17⅝	17¾
7. Crotch height—do	29⅝	29¼	29⅛
8. Bitrochanteric diameter—do	12	12⅝	13⅛
9. Shoulder slope—degrees	25	25	24½
10. Anterior chest width—inches	11½	11⅞	12⅛
11. Anterior waist length—do	12⅜	12⅝	12¾
12. Posterior chest width—do	12¾	13	13¼
13. Posterior waist length—do	14¼	14⅜	14⅜
14. Chest girth at armscye—do	30¾	32	33⅛
15. Scye depth—do	6⅞	7	7¼
16. Posterior hip arc—do	16½	17½	18½
17. Maximum chest girth—do	31¼	32⅝	33⅞
18. Anterior chest arc—do	16⅜	17⅛	17¾
19. Waist girth—do	24⅜	25⅛	26⅛
20. Hip girth—do	33⅝	35⅜	37¼
21. Neck-base girth—do	14	14¼	14½
22. Shoulder length—do	4⅜	4⅜	4⅜
23. Armscye girth—do	14⅛	14⅝	15
24. Upper-arm girth—do	8¾	9⅜	10
25. Elbow girth—do	9⅜	9⅞	10¼
26. Upper posterior arm length—do	13¼	13¼	13⅜
27. Total posterior arm length—do	23⅛	23¼	23⅜
28. Trunk line—do	7½	7⅝	7½
29. Waist to hips—do	8⅛	8⅜	8½
30. Thigh girth—do	19	20⅜	21¾
31. Maximum calf girth—do	12¼	12⅞	13⅜
32. Knee girth—do	12⅞	13¼	13⅜
33. Total crotch length—do	26⅜	27¼	28⅛
34. Anterior crotch length—do	13	13½	13⅞
35. Extreme bend—do	28	28⅜	28⅞
36. Vertical trunk girth—do	54½	55⅞	57
37. Percentage [1]—percent	21	38	24

Item No. and measurement	B_{12}	C_{12}	D_{12}
	Height, 66; hip, 35	Height, 66; hip, 36½	Height, 66; hip, 38½
1. Waist height—inches	42⅛	42⅛	42¼
2. Hip height—do	34⅛	34	34
3. Weight—pounds	112	122½	134
4. Stature—inches	66	66⅛	66⅛
5. Cervicale height—do	56½	56⅝	56¾
6. Tibiale height—do	18⅜	18⅜	18⅜
7. Crotch height—do	30⅝	30⅜	30¼
8. Bitrochanteric diameter—do	12½	13	13½
9. Shoulder slope—degrees	25½	25	24½
10. Anterior chest width—inches	11⅞	12⅛	12½
11. Anterior waist length—do	12⅞	13	13⅛
12. Posterior chest width—do	13⅛	13⅜	13⅝
13. Posterior waist length—do	14⅜	14⅞	14⅞
14. Chest girth at armscye—do	31½	32¾	34
15. Scye depth—do	7⅛	7¼	7½
16. Posterior hip arc—do	17⅛	18	19
17. Maximum chest girth—do	32⅛	33⅜	34⅝
18. Anterior chest arc—do	16¾	17½	18⅛
19. Waist girth—do	24¾	25⅝	26¾
20. Hip girth—do	34¾	36⅝	38½
21. Neck-base girth—do	14⅜	14⅝	14¾
22. Shoulder length—do	4½	4½	4½
23. Armscye girth—do	14½	15	15½
24. Upper-arm girth—do	9	9⅝	10¼
25. Elbow girth—do	9⅞	10¼	10½
26. Upper posterior arm length—do	13⅝	13¾	13⅞
27. Total posterior arm length—do	24	24⅛	24¼
28. Trunk line—do	7⅞	7¾	7¾
29. Waist to hips—do	8½	8⅝	8⅞
30. Thigh girth—do	19¾	21⅛	22⅜
31. Maximum calf girth—do	12⅝	13¼	13¾
32. Knee girth—do	13⅛	13⅝	14⅛
33. Total crotch length—do	27⅜	28⅛	29
34. Anterior crotch length—do	13½	13⅞	14¼
35. Extreme bend—do	29	29⅜	29⅞
36. Vertical trunk girth—do	56¼	57½	58⅝
37. Percentage [1]—percent	25	38	22

[1] Based on total numher of children in the given height interval.

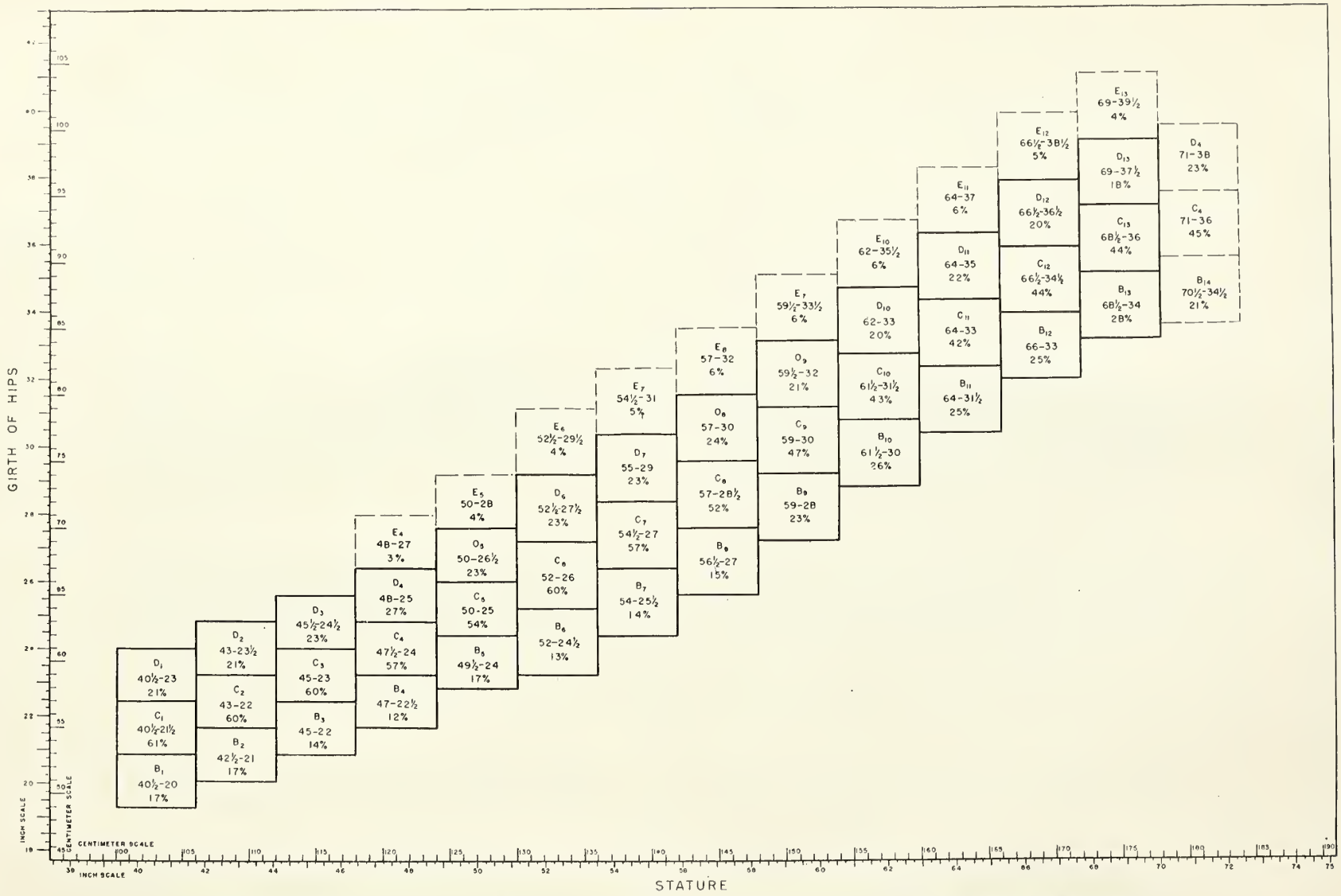


FIGURE 3.—System of boys' sizes based on stature and girth of hips following the distribution shown in figure 1. The symbols, B_1, C_1, D_1, and others in the sequence, designating rectangles outlined by unbroken lines, refer to the "below regular," "regular," and "above regular" sizes as given in table 8. Rectangles outlined by dotted lines show sizes for children whose proportions vary from those of the majority. The percentages are based on the total number of children in a given height interval.

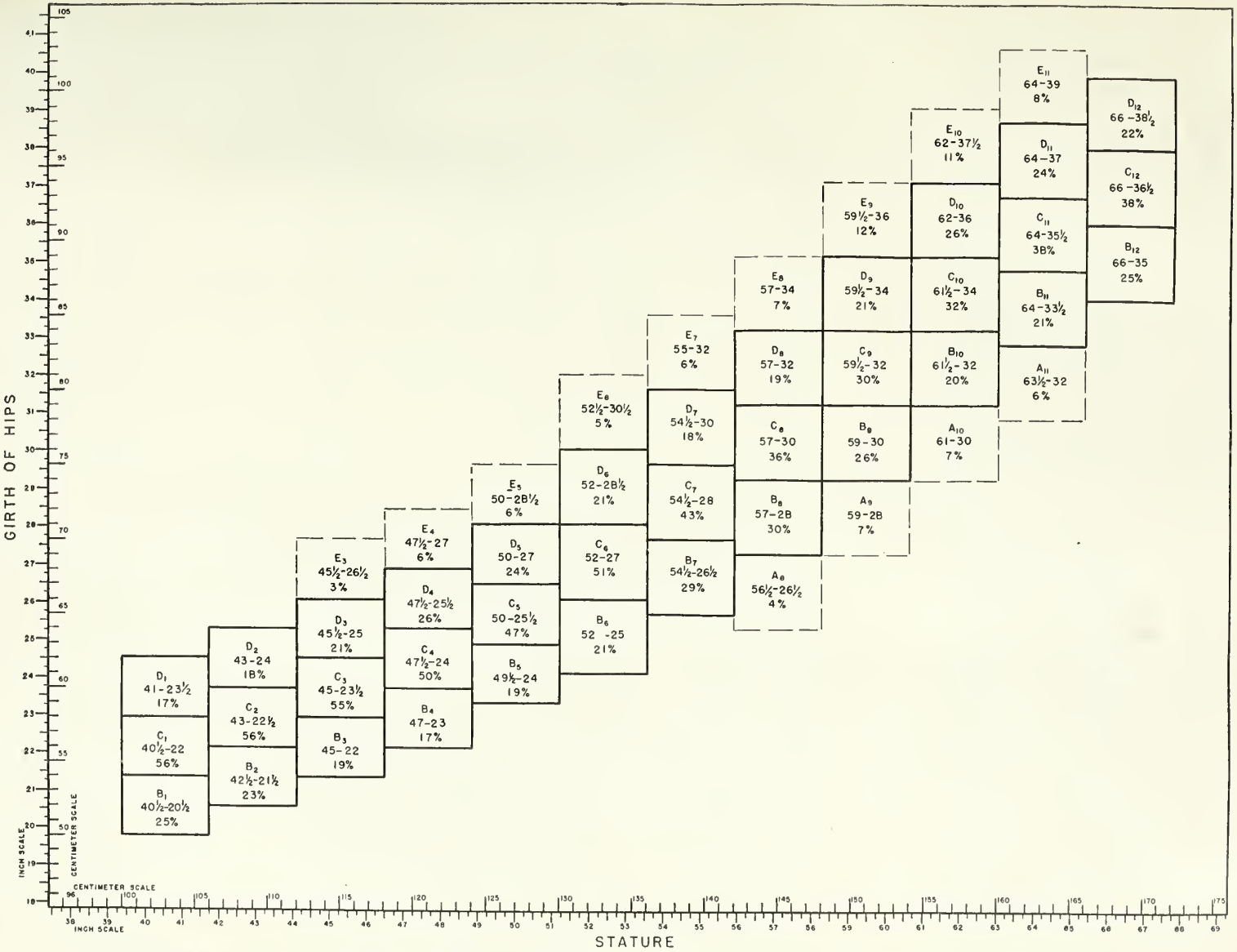


FIGURE 4.—System of girls' sizes based on stature and girth of hips following the distribution shown in figure 2. The symbols, B_1, C_1, D_1, and others in the sequence, designating rectangles outlined by unbroken lines, refer to the "below regular," "regular," and "above regular" sizes as given in table 9. Rectangles outlined by dotted lines show sizes for children whose proportions vary from those of the majority. The percentages are based on the total number of children in a given height interval.

Proposed System not Based on Age

The proposed system is based on stature and hip measure and has no relation to age sizes.

Since children 4 to 17 years of age were included, the system applies to children of these ages. However, many 2- and 3-year-olds will undoubtedly be included in the small sizes, and children older than 17 will be included in the large ones. This depends on their height and their hip measure. Similarly some 4- and 5-year-olds will be too small for these sizes, and some 16- and 17-year-olds will be too large. In other words, boys who have the height and hip combinations that come within the range of 39½ inches and 70 inches for height and 19¼ inches and 39 inches for hip measure and girls who have the height and hip combinations which come within the range of 39½ inches and 67¾ inches for height and 19¾ inches and 40 inches for hip measure will be included regardless of their age. Those who are shorter or taller or who have smaller or larger hips than are represented in these ranges are not considered in this scheme.

APPENDIX

Measurements and Methods of Taking Them

From the technical report submitted *by* Eleanor P. Hunt, *associate anthropometrist*

The measurements listed below are included in the proposed standard system of body measurements.

1. Height of waist.
2. Height of hips.
3. Weight.
4. Stature.
5. Height of cervicale.
6. Height of tibiale.
7. Height of crotch.
8. Bitrochanteric diameter.
9. Slope of shoulder, right.
10. Width of chest, front.
11. Length of waist, front.
12. Width of chest, back.
13. Length of waist, back.
14. Girth of the chest at the armscye.
15. Depth of scye.
16. Back arc of hips.
17. Maximum girth of chest.
18. Front arc of the chest.
19. Girth of waist.
20. Girth of the hips.
21. Girth of the neck base.
22. Shoulder length, right.
23. Girth of the armscye, right.
24. Girth of the upper arm, right.
25. Girth of the elbow, right.
26. Length of right arm, upper segment, back surface.
27. Total length of right arm, back.
28. Trunk line, right.
29. Length from waist to hip, right.
30. Maximum girth of the thigh, right.
31. Maximum girth of calf, right.
32. Girth of the knee at tibiale right.
33. Length of crotch, total.
34. Length of the crotch, front.
35. Extreme bend.
36. Trunk girth, vertical.

The instruments used in the study consisted of a calibrated anthropometer and steel tape similar to standard anthropometric instruments, a pair of calipers for making bisections of distances, and an instrument for measuring shoulder slope which was devised in the Bureau. A skin pencil was supplied for placing the landmarks on the body, a steel knitting needle to use as a ruler in establishing needed vertical lines, and a small chain to outline the neck base. All measurements except weight and shoulder slope were taken in the metric system.

The landmarks described below are placed on the body with a skin pencil and used later as reference points for the measurements. They are grouped here in the following order: Neck, trunk, arm, hip, and leg, although in the study, a different order was used which permitted greater speed in the routine measuring.

Placing the Landmarks

The neck base.—A fine-gage, flexible-link chain is looped around the neck so that it touches the upper border of the medial ends of the clavicles (fig. 5) and rests on the cervicale (fig. 6). While the chain is in place its position is marked in the center front and over the trapezius muscle on the right and left side. A short vertical line intersecting the neck base is drawn at the center front. A cross is placed at the cervicale on the prominence of the spinous process of the seventh cervical vertebra. This prominence is found more readily when the head is forward, but the landmark is made when the skin is in normal position and the head erect.

The armscyes.—The position of the seam of a set-in sleeve is marked on the right and left arms by the use of four landmarks. The determining points are the shoulder, the armscye back and front, and the underarm midpoint. The procedure of locating the landmarks is identical for the right and left sides and will be described for the right side only.

The shoulder point desired is midway between the acromion and the highest point at the lateral end of the clavicle (fig. 6).

In this study the point marked as the acromion is on the side of the acromial process midway between a point in front of the angle of the process and a point at the center of the shoulder, as judged by sighting. The first step is to outline the side of the acromial process. Points are then placed on the outline at the angle of the process and at the sighted center of the shoulder. A line intersecting the outline of the process is placed midway between these two points, and the intersection marked as the acromion.

This position corresponds very closely to the acromion as defined anatomically as the most lateral point of the margin of the acromial process. The use of the midpoint between these easily located limits makes it easier to locate the most lateral point of the process. The highest

point at the lateral end of the clavicle is then found and marked. When the acromion and the highest point of the side of the clavicle have thus been placed, the shoulder point of the armscye is located midway between these points.

The back and front of the armscye are traced with the aid of a chalked string. The center of the string is placed under the arm when the arm is raised about 30° from the trunk. The ends of the string are raised and crossed up over the shoulder point, thus indicating the direction of the armscye at the back and front, as well as under the arm (figs. 5 and 6). The chalked path of the armscye is marked with the skin pencil in short, thin, sloping lines.

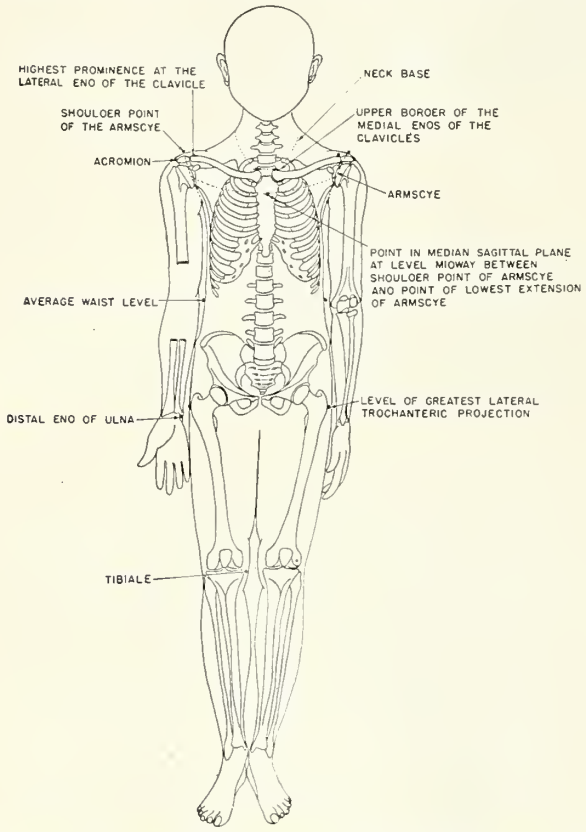


FIGURE 5.—Body landmarks used in taking measurements. Front view.

These lines are placed on the front and back of each shoulder and on the top of the right shoulder but not under the arm.

The underarm midpoint is located with reference to the natural folds in the armpit and the total width of the shoulder. The height of the midpoint in the armpit is decided on the basis of the size and position of the folds. The observer sits in front of the subject and observes the formation of the folds on the right and left sides as the arms are raised to about 45° and gradually lowered to meet the trunk. Usually the folds of one armpit are more clearly defined than those of the other. The level of the midpoint is set with respect to the more clearly defined folds, and a corresponding level is marked off in the pit of the arm on the opposite side. A short, thin, slightly curved line on the trunk marks the underarmscye level.

A vertical line indicating half of the total width of the shoulder is drawn through this to give the underarm midpoint of the trunk. This bisection of the shoulder is made with a small caliper. The upper edge of the shaft of the caliper is held against the trunk at the level of the armscye. The shaft is horizontal, and the jaws are in a vertical position. The jaws touch the shoulder at the back and front without constricting it. The subject's arm is raised to the side at an angle of approximately 90° with as little elevation of the shoulder as possible. With the caliper in this position, the midpoint of the total width is marked with the skin pencil.

The position of the underarm midpoints on the arm as well as on the trunk should take clothing construction into account. The measurer must decide to what height under

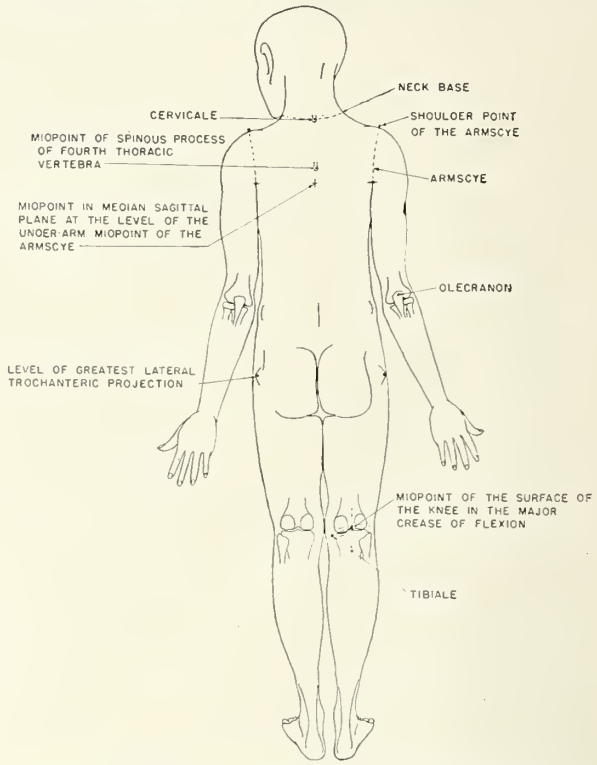


FIGURE 6.—Body landmarks used in taking measurements. Back view.

the arm the blouse can extend without forming an uncomfortable surplus of fabric when the arm is lowered. This depends on the position of the armpit folds on the subject being measured. The highest level considered feasible is the one chosen.

The shoulder line.—This line intersects the neck base and the armscye at the shoulder. It corresponds to the customary shoulder seam of a garment and is located with reference to the trapezius muscle and the acromion. By feeling at the neck base, the border of the upper fibers of the trapezius which pass forward and downward to become inserted in the acromial end of the clavicle will be found. The intersection of the shoulder line with the neck base is placed at the front border of the trapezius. The other end of the shoulder line is directed by the acromion, although the intersection of the shoulder line is actually with the

armscye. A small steel knitting needle may he used to guide the observer so that the neck base and armscye intersections can readily he placed with respect to the trapezius border and the acromion.

Average waist level.—The waist level used lies at the lower edge of the lowest rih and is found hy feeling the sides of the body in line with the armpit. This waist level corresponds very closely to the natural waist which can he seen when the side profiles of the body are slightly concave. A natural waist in this sense does not often occur among young children, but a waist level hased on the lower edge of the lowest rib found hy feeling at the side of the hody can be used equally well in all age groups. This waist level also provides for the maximum depth of a garment from the waist to the crotch level. In this respect, it is preferable to a waist placed at the crest of the ilium, a level which is readily determined, hut results in a waist of minimum height and in shallow crotch measurements.

To locate the waist, the observer sits in front of the subject and feels the right and left sides simultaneously, using the index fingers to press against the sides in line with the armpits. The hands are held with the palms directed toward the floor. The fingers are extended and together. The thumb side of the middle joint of the index finger is placed against the suhject. When the lower edge of the lowest rib is felt on the hack surface of the index finger, the level of the midline of the index finger is taken as the waist level. Without displacing the skin, the level is marked with a point in line with the armpit on the right and left sides. The waist levels of the right and left sides frequently differ. The average height from the floor of the two sides is considered the waist level. The anthropometer is used to find the average height at the center front, center back, and at the right side of the back. If the difference in height between the right and left sides exceeds 4 millimeters, the points first placed on the sides are corrected to correspond with the average height of waist.

The finished landmarks of the waist are five in number: center back, center front, each side, and the right side of the back. Small crosses composed of two short, straight, thin lines at right angles, similar to those at the neck hase and cervicale are used. The horizontal hranches indicate average height of waist. The intersection at the right side of the back is located hy bisecting the distance measured with the tape between the center back and the center side.

Levels for the measurement of width of chest.—Short, straight, thin lines are used to indicate the levels on the back and chest at which the width of chest hetween the armscyes is measured. On the hack, the landmark is placed in the center on the prominence of the spinous process of the fourth thorasic vertebra. The landmark on the chest is placed at a level midway hetween that of the shoulder point of the armscye and the level of the lowest visible point of the armscye. The latter point is indicated by placing a pencil under the arm so that the hlunt end is visihle at the juncture of the arm and trunk.

Level for the measurement of maximum chest girth.—To guide the measurer in placing his tape at the level at which the girth of chest appears to he the greatest, a landmark is placed at this level after ohserving carefully the outline of the chest. The region of the chest is viewed from one side, and the hack and front profiles of the hody are considered. A preliminary landmark is placed at the level at which it appears that the girth will be the greatest. This may he placed on the back or the front according to convenience. Similarly, preliminary landmarks are placed when the suhject is viewed from the opposite side. The average height of the two preliminary landmarks is taken at the level of maximum chest girth. The final landmark may be placed in the center back or, if the front profile has heen used to set preliminary landmarks, it may he more convenient to place the final landmark on the front.

On hoys younger than 11 years of age, these landmarks are not used, and the measurement of maximum girth of chest is omitted. Landmarks and measurements are also omitted on girls younger than 10 years. Among the girls in the older age groups, the level corresponds to the hust measurement.

Elbow, right.—The length of the upper segment of the arm is measured from the armscye–shoulder line intersec-

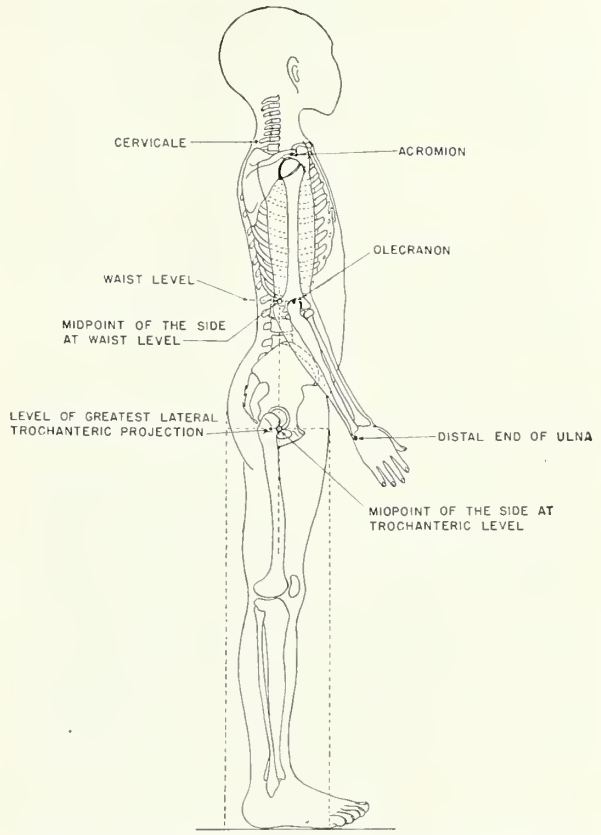


FIGURE 7.—Body landmark used in taking measurements. Side view.

tion to the elbow. The elbow is marked at the point farthest to the side when the closed fist is rested against the waist with the back of the hand facing the front.

Wrist, right.—The total length of the arm is measured from the armscye–shoulder line intersection over the elbow to the farther (distal) end of the ulna at the wrist (figs. 5 and 7). To locate the end of the ulna, the flat of the thumbnail of the observer is pushed upward against the end of the ulna on the side corresponding to the littlc finger. A short line is drawn perpendicular to the long axis of the ulna. An intersecting line is placed at right angles to this in the middle of the side of thc wrist corresponding to the little finger.

Average hip level.—The level of the hip is placed at the most prominent bony point in the region of the trochanter major. The level is determined independently for the right and left sides. The levels of the two sides usually differ, and the average of the two is taken as the hip level. The preliminary landmarks are corrected to correspond with

the average if the difference between the two sides exceeds 1 mm.

The extended index and middle fingers of the observer's right hand are used to feel the region of the trochanter. This is done while the observer squats with eyes approximately at the level of the trochanter. The direction of palpation is from below upward. On well-developed, muscular individuals and when excessive fat pads are present, it will take some time to find the proper level. The subject may be asked to bend slightly forward or rotate the femur by turning the toes laterally and by pivoting on the heel. A rounded region is felt rather than a point. The midpoint of this region is marked with a preliminary point to indicate the hip level. The average hip level is marked with a short horizontal line.

A point on this horizontal line corresponding to the location of the side seam of a garment is obtained by sighting between the greatest projection of the buttocks and the front projection of the thigh. This is done by the observer squatting at the side of the subject with eyes at the hip level, holding the knitting needle and skin pencil tangent to the buttocks and thigh profiles, respectively, at their most projected parts. The midpoint between these two is determined and marked on the hip-level line. When this point is set, the knitting needle is held vertically and used as a ruler to extend a line from this point to the waist level. The point where it meets the waist level is marked (fig. 7).

The crotch center corresponds to the intersection of the inside seams of trouser legs with the crotch seams. The subject rests his left foot on a chair while this landmark is placed. The observer squats at the left side of the subject with his eyes at the level of the fold of the left buttock and directly in front of the inner surface of the right thigh. The subject's right foot is placed straight forward. The middle line of the inner surface of the right thigh is sighted and marked with a thin vertical line extending downward from the level of the perineum. If the lower borders of the measuring suit obscure the region, the subject is asked to draw them up by lifting the suit by the waist band.

Knee center at back.—The height of this landmark is determined with reference to the natural creases of the skin at the back of the knee. By flexing the subject's knee, the creases can be clearly seen. The direction of the most pronounced crease is outlined. A vertical intersection is drawn in the middle line of the back region of the knee.

Tibiale, right.—In this study the middle of the "cleft" between the bones that meet at the knee has been used as a guide. The cleft is located by grasping the knee firmly while it is alternately flexed and extended and by moving the index finger or thumb in the region of the cleft with pressure until the margins of the bones are found. The tibiale is marked by a dot as shown in figure 6 at the level of the cleft which corresponds to the highest point which can be felt at the end of the tibia. In practice the midlevel of the cleft gives a close approximation of the tibiale.

Procedure Followed in Taking Measurements

After all the landmarks have been penciled on the body the measurements are taken as described below.

1. *Height of waist.*—This measurement is the average height of the preliminary landmarks set at the waist. The subject stands erect facing the observer with feet together. His weight is evenly distributed between the two feet and he is cautioned against shifting his weight from one foot to the other and from heels to toes and vice versa. His arms hang loosely at the side, somewhat toward the back. The observer sits in front of the subject or stands when the height of waist equals or exceeds 1 m. The measurement is taken with the anthropometer, and the leveling platform is used if the floor of the workroom is not level.

2. *Height of hips.*—This is the average height of the preliminary landmarks placed on the right and left sides in the region of the trochanters and is taken with the anthropometer, the leveling platform being used if the floor of the workroom is not level. The subject stands facing the observer with hands on the hips, feet together, and weight evenly distributed. The observer ordinarily finds it convenient to sit or squat in front of the subject.

3. *Weight.*—The subject is asked to stand quietly on the center of the platform of the scale with hands on the hips. He is instructed not to shift his body while the reading is made.

4. *Stature.*—The subject stands on the leveling platform with heels against the wall and together. The shoulders and buttocks just touch the wall. The eyes are directed forward, and the head is erect. The palms of the hands lie on the thigh. The observer stands at the subject's right side. The measurement is taken with the anthropometer, and a perpendicular wallboard is used when a suitable wall, free of baseboard and paneling, is not available in the workroom.

The anthropometer is held and balanced in a vertical position in the right hand. The left hand locates the top of the head while the right hand slides the moving arm of the anthropometer down to rest there. Sufficient pressure is used to bring the point of the anthropometer to the level of the top of the head.

5. *Height of cervicale.*—The subject's position is identical with that described for the measurement of stature, except that the subject stands away from the wall toward the outer end of the leveling platform so that the anthropometer can be placed back of him. The observer stands back of the subject and to his left side. The anthropometer is held vertical in the right hand in line with the center back. The straight edge of the brass point piece is directed toward the floor. The point is lowered to the cervicale. The measurement is taken quickly before the child alters his position. The anthropometer is used for this measurement. The leveling platform will be required if the floor of the workroom is not level.

6. *Height of tibiale.*—The upper section of the anthropometer is used. If the floor of the workroom is not level, the subject stands on the leveling platform. He places his left foot on a chair which raises the foot about to the level of the midregion of the right kneecap. The subject's weight is evenly distributed between the two feet. So far as possible, the main axis of the right leg is perpendicular to the floor. The right foot is directed straight forward. The observer squats at the subject's left side with eyes at the knee level. The upper section of the anthropometer is inverted so that it stands on the fixed arm. The straight edge of the brass point piece is directed toward the floor. The point is raised to the level of the tibiale. The reading is taken at the upper margin of the brass collar of the movable horizontal arm of the anthropometer.

7. *Height of crotch.*—The upper section of the anthropometer is used; but if the floor of the workroom is not level, the leveling platform is used. The subject's position is the same as for the measurement of the height of tibiale. Before the left foot is raised to the chair, the subject is asked to pull up the measuring suit by the waist band in order to bring the garment in snug contact with the perineal floor. The principal transverse axis of the pelvis is approximately horizontal to the floor. The observer

squats at the center back of the subject with eyes at the level of the fold of the buttocks.

The anthropometer rests on the fixed arm. The brass point piece is raised in the crotch until its straight edge is at the level of the base of the left buttock. This level is indicated by placing a pencil under the left buttock, tangent to the buttock at its lowest point and parallel to the floor. The straight edge of the brass point piece is brought to rest on the pencil at this level of the left buttock. Caution is exercised in order not to tip the pencil from its horizontal position, nor exert more than moderate pressure on the under surface of the buttock.

8. *Bitrochanteric diameter.*—The upper section of the anthropometer is used for this measurement. The subject stands as for the measurement of stature except that he is away from the wall and hands are placed on the hips. The weight is distributed equally on both feet. It is usually convenient for the observer to sit in front of the subject.

The straight edge of the brass point piece faces the fixed arm of the anthropometer. The shaft and arms of the anthropometer lie in a plane parallel to the floor. The midpoints of the inner surfaces of the arm are placed on the hip landmarks. The anthropometer is closed without pressure on the skin and the reading made at the inner margin of the brass collar of the movable arm of the instrument.

9. *Slope of shoulder, right.*—The subject stands in the same position as for the measurement of the cervicale, with arms relaxed and hanging at the sides. The observer stands in back of the subject and to his right side. The wooden blade of the protractor is rested on the intersection of the shoulder line and the right armscye. The subject is asked to carry the weight of the instrument on his shoulder without altering the position of the shoulder.

The intersection of the shoulder line and the armscye is used as the pivotal point, and the opposite end of the wooden blade of the protractor is lowered to the point where the shoulder line and the neck base intersect. The instrument should rest lightly without depressing the skin. It is balanced easily on the landmark while the left hand is used to steady the instrument and the right thumb and index fingers adjust the spirit level to a horizontal position. If the protractor cannot be brought to rest on the intersection of the shoulder line and the armscye because of the interference of higher points, the blade is poised on the highest point; but the blade's direction is that of the shoulder line.

10. *Width of chest, front.*—The subject's position is the same as that taken for the measurement of stature except that he stands away from the wall. The observer is in front of the subject with eyes at the level of the measurement. Sitting or squatting may be necessary when measuring the younger groups.

The observer tells the subject that this measurement and the three following immediately are to be made while he stands in the same position. It is essential that the subject's position corresponds with the standard position for stature and that the position be maintained throughout the set of four measurements. The upper border of the tape is rested at the level of the landmark previously made on the front of the chest. The measurement is made from the line marking the armscye on one side to the outline of the other armscye. The upper border of the tape is held horizontal. It may be necessary to elevate the zero and reading points of the tape in order to bring its upper edge into a horizontal position.

11. *Length of waist, front.*—The subject's position is identical with that described in the preceding section. The zero point of the tape rests at the landmark made at the neck base, and the reading is made at the average waist level in the center front, previously marked.

12. *Width of chest, back.*—The subject's position is identical to that taken for the measurement of width of chest, front. The observer is back of the subject, standing, sitting, or squatting so that his eyes are at the level of the measurement. The upper border of the tape rests at the level of the landmark made on the spinous process of the fourth thoracic vertebra and is made to lie in a horizontal plane by elevating the zero and reading points of the tape as needed. The measurement is taken without constriction and is made from the line marking the armscye on one side to the outline of the other armscye.

13. *Length of waist, back.*—The position of the subject is identical to that described for the measurement of the width of chest, front. The observer is back of the subject and to the left side. The zero point of the tape is placed at the landmark at cervicale. The tape follows the outline of the center back without constriction, and the reading is made at the landmark at the average waist level in the center back.

14. *Girth of the chest at the armscye.*—The subject's position is his normal erect posture. The observer stands back of the subject. The tape is placed around the trunk without constriction, with the zero point at the center back. The upper border of the tape rests under the arms at the level of the landmark of the armscye on the trunk and passes through the underarm midpoints of the left and right armscyes, as previously marked. When the subject's arms are raised slightly in order to place the tape in proper position in relation to the underarm midpoints, the tape should be somewhat slack in the front. If this precaution is observed, the tape will not constrict the chest muscles when the arms are lowered. If the subject elevates his shoulders, he should be asked to relax them.

When the tape has been placed so that its upper border passes through the landmarks of the underarm midpoints and the girth is set without constriction, the observer passes around to the side of the subject to check the relative positions of the back and front arcs of the girth. The two arcs should lie horizontal to the floor. It is frequently necessary to lower the arcs and set the tape more loosely. The subject breathes normally and the middle value between the largest and smallest reading is recorded. Before removing the tape the level of its upper border is marked in the back by a dot in the center of the body. In this way the approximate level of the underarm midpoints is taken on the back and a landmark is provided for the measurement to follow.

15. *Depth of scye.*—The subject assumes a normal posture with head erect and eyes directed forward. The observer stands at the center back of the subject. The zero point of the tape is placed on the landmark at cervicale. The subject should be cautioned not to lower his head. The reading is made at the landmark placed in the center of the body during measurement No. 14. This landmark gives the position of the border of the tape when the latter was placed for the measurement of the girth of chest at the armscye.

16. *Back arc of hips.*—The subject stands erect with feet together, weight evenly distributed, and hands on the hips. The observer squats back of the subject with eyes at hip level. The zero point of the tape is placed at the hip level of the left side. The upper border of the tape lies in a horizontal plane. The reading is made on the right thigh at the hip level, the measurement being taken without constriction.

17. *Maximum girth of chest.*—The subject's position is his normal erect posture with feet together. The tape is passed around the chest so that the upper border is at the level previously marked for the measurement of the maximum girth. The zero point is at the center front or center back depending upon the position of the landmarks.

(See p. 21.) The front and back arcs of the girth should be horizontal to the floor. The girth is measured without constriction. This measurement is omitted on boys younger than 11 years of age and girls younger than 10. If the level of the girth of chest at the armscye as marked is the same as that indicated for the maximum girth, the reading for the girth of chest at the armscye is recorded for the maximum girth of chest as well.

18. *Front arc of the chest.*—The subject's initial position is his normal erect posture with feet together, and arms relaxed at the side. The observer should raise the arms of the subject just enough to permit the observer to see the underarm midpoints. The minimum movement to expose these points and yet not disturb their position can be accomplished if the shoulder is not disturbed and the movement of the shoulder is confined to the elevation of the arm. The arms are slightly bent at the elbow. The hands are placed far enough from the side to permit the observer to work between the arms and the trunk. Care is taken that the shoulders are not thrown forward giving the subject a hollow chest appearance and thus reducing the accuracy of the reading. The observer is in front of the subject with eyes at the level of the chest. The zero point of the tape is placed on the landmark at the right underarm midpoint. The upper border of the tape lies horizontal to the floor, and the reading is made at the landmark at the left underarm midpoint.

19. *Girth of the waist.*—The subject's position is his normal erect posture with feet together. The observer is in front of the subject. The tape is placed around the body at waist level with the upper border of the tape passing through the landmarks at the average waist level. The zero point is at the center front. The measurement is taken without constriction. The subject breathes normally. The middle value between the largest and smallest reading is recorded.

20. *Girth of the hips.*—The subject's position is his normal erect posture with feet together and hands on the hips. The observer is at the right side of the subject. When the reading is made the eyes are at hip level. The tape is passed around the body so that the upper border passes through the landmarks made for the average hip level. The tape should be held horizontal to the floor. The measurement is taken without constriction.

21. *Girth of the neck base.*—The subject's position is his normal erect posture. The observer stands to the right side of the subject. The tape is placed around the neck with its lower border passing through the landmark placed on the cervicale. The zero point of the tape lies just back of the intersection of the neck base with the shoulder line. The thin edge of the tape rests on the markings of the neck base of the right and left sides and the center front. The observer fits the tape to the landmarks in succession, releasing each placement after the one following is completed. The tape should describe a smooth, closed curve through the four landmarks. The measurement is taken without constriction and without undue slack.

22. *Shoulder length.*—The subject's position is his normal erect posture with feet together and arms relaxed at the sides. The observer stands back of the subject and to his right side. The zero point of the tape is placed at the landmark where the neck base intersects the shoulder line. The top border of the tape is placed on the shoulder line. The subject should be cautioned against lowering his shoulder and drawing the head away from the tape at its zero position.

23. *Girth of the armscye, right.*—The subject's position is his normal erect posture with feet together and arms relaxed at the sides. The observer is back of the subject and to his right side. The subject's arm is raised sufficiently to permit the observer to place the tape under the arm so that the thin edge passes through the underarm midpoint and the zero point falls just below the armscye at the back. The case end of the tape is brought up over the shoulder, and the case is allowed to drop over the back, thus providing the tension for this measurement. The folds of skin at the front of the armpit are eased under the tape. The tape is fitted to the back and front of the armscye as well as to the landmark where the armscye intersects the shoulder line. The looped end of the tape is raised without unduly binding the skin folds at the back of the armpit.

24. *Girth of the upper arm, right.*—The subject's position is his normal erect posture with feet together and arms relaxed at the sides. The observer is at the right side of the subject with eyes at the level of measurement. The tape is placed around the upper arm. The girth is taken midway between the shoulder and the elbow without constriction. The tape should be horizontal when the arm hangs relaxed so that the principal long axis of the arm is approximately perpendicular to the floor.

25. *Girth of the elbow, right.*—The subject stands in a normal, erect position. The right arm is bent at the elbow, and the hand and fingers extended forward. The angle of the elbow is approximately 90° with the upper arm directed straight downward. The observer is at the right side of the subject and with eyes at the level of the measurement. The tape is placed around the arm so that it passes over the landmark made on the elbow and bisects the angle of the bent elbow. The zero point is in front of the observer. The measurement is taken without constriction.

26. *Length of right arm, upper segment, back surface.*—The subject stands in a normal erect position with feet together. The right fist is placed on the hip, with the back of the hand toward the front of the body. The arm is not bent at the wrist. The observer stands to the back of the subject, at his right side. The zero point of the tape is placed at the point where the armscye intersects the shoulder line. The reading is made at the landmark previously placed at the elbow. The tape is continued over the elbow to the wrist.

27. *Total length of right arm, back.*—The subject's position is identical to that assumed for measurement No. 26. The observer stands to the right side and back of the subject. When the reading is made, the observer bends down under the point at the wrist so that the reading is made in the direct line of vision. The zero point of the tape is placed on the landmark where the armscye intersects the shoulder line. The tape passes over the elbow to the landmark at the distal end of the ulna, at the wrist (fig. 7). The measurement is taken with sufficient tension to maintain the tape in position at the elbow.

28. *Trunk line, right.*—The subject's position is normal erect posture with feet together. The observer asks the subject to relax his right arm and shoulder and permit him to place the arm in the desired position. The forearm is bent at an angle of 90° to the upper arm, and the hand is extended directly forward. While the observer steadies the subject's shoulder, he moves the entire arm thus bent, an inch or so directly forward. The observer, still steadying the shoulder, then grasps the bent elbow and tips it slightly upward until the underarm midpoint is just visible when the observer's eyes are at the level of the midtrunk. The object of this detailed procedure is to expose the underarm midpoint without moving it significantly from its position when the arm hangs loosely at the side. The observer sits at the right side of the subject with eyes at the level of the middle region of the trunk. The zero point of the tape is placed at the underarm midpoint. The tape passes directly to the landmark placed at the average waist level. If the subject is wearing a handeau the tape may pass over or under the garment, whichever will disturb less the direct course of the tape between the landmarks.

29. *Length from waist to hip, right.*—The position of the subject and observer in taking measurement No. 28 is maintained. The zero point of the tape is placed at the landmark of the average waist level. The tape passes over the measuring garment, and the reading is made at the landmark placed at the average hip level.

30. *Maximum girth of the thigh, right.*—The subject's position is his normal erect posture with hands on the hips. The feet are parted slightly to permit the tape to pass freely between the thighs. The observer is at the subject's right side with eyes at the level of the folds of the buttocks. The tape, which is placed around the thigh with the upper border at the level of the fold of the buttock, is horizontal to the floor. The zero point is situated directly in front of the observer at the right side of the subject's thigh. The observer passes to the front of the subject after the tape is placed to see that the tape is in the proper position. If the fold of the buttock is not distinguishable, it can be found by pressure on the right buttock. However, several folds may be so produced and the major one of these must be selected provisionally. The tape passes over the thigh at its largest point. The measurement is taken without constricting the thigh.

31. *Maximum girth of calf, right.*—The subject's position for measurement No. 30 is maintained unaltered. The observer squats at the right side of the subject with eyes at the midlevel of the tibia. The tape passes around the calf at the level where the observer judges the girth to be maximum. The level is determined by looking at the back profile of the calf. The tape is held horizontal to the floor, and the measurement is taken without constriction of the calf.

32. *Girth of the knee at tibiale, right.*—The subject's position is the same as that described for measurement No. 31. The observer squats in front of the subject with eyes at the level of the tibiale. The tape is placed around the knee with the upper border at the level of the tibiale. The zero point is at the center front. The tape is held horizontal to the floor. The measurement is taken without constricting the knee.

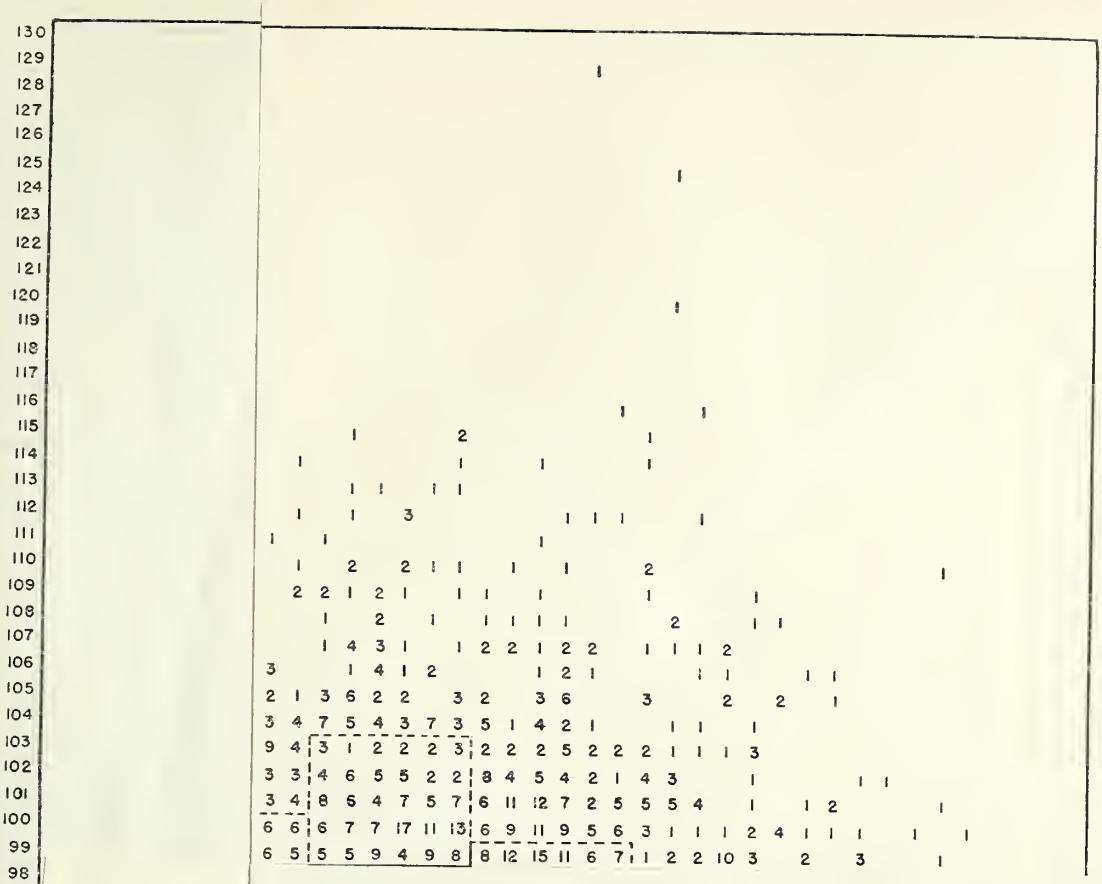
33. *Length of crotch, total.*—The subject is asked to pull up the measuring suit by the waist band to insure a snug adjustment of the suit at the crotch. The left foot is placed on a stool or other steady object which elevates the left foot to the level of the middle of the right kneecap. The weight is evenly distributed between the two feet. The long axis of the right leg is approximately perpendicular. The right foot is directed straight forward. The left hand rests on the left thigh. The head and trunk are erect. The principal transverse axis of the pelvis is approximately horizontal. The observer is at the left side of the subject while placing the zero point. He is at the center back with eyes at average waist level when the reading is made.

The zero point of the tape is placed at the landmark on the average waist level in the front. The subject holds the tape in position with two fingers flat against the abdomen and spread so that the position of the zero point can be checked without moving the fingers. On very young children the observer holds the zero point of the tape in position. The tape passes centrally over the genitals and floor of the crotch to the landmark of the average waist level in the back. The tension of the tape is about that provided by the weight of the tape case.

34. *Length of the crotch, front.*—The subject's position taken for measurement No. 33 is maintained. The observer squats at the subject's left side with eyes at crotch level. The zero point of the tape is placed at the landmark at the average waist level in front. When the reading of the total length of crotch has been made (measurement No. 33), the observer drops the tape from the average waist level in the back to the crotch level. There the tape is supported on the observer's left index finger, and the case hangs free, providing the tension desired for this measurement. The tape is lowered from the floor of the crotch by the width of the tip of the observer's index finger so that the observer can grasp the tape between the index finger and thumb at the middle landmark of the thigh. This landmark is the reading point for measurement.

35. *Extreme bend.*—The subject stands with feet together facing a chair or stool about 2 feet away. He bends over at the hips and grasps the leg of the chair, or otherwise brings his hands to rest so that they are approximately at the middle or lower border of the kneecap. The head is allowed to drop forward. The legs are straight and perpendicular to the floor. It is necessary to adjust the position of the chair so that the legs and arms can be brought to the desired position without discomfort for the subject. The observer is back of the subject. He squats with eyes at knee level while the direction of the tape is checked and the reading is made. The zero point of the tape is placed at the point of the average waist level midway between center back and right side. The tape passes without constriction over the buttocks to the midpoint of the knee at the back. Throughout its course the tape is equidistant from the center line of the body; that is to say, the arc of the tape, although fixed at the landmarks, does not swing to either side or to the middle. The reading is made at the landmark midpoint of the knee in the back.

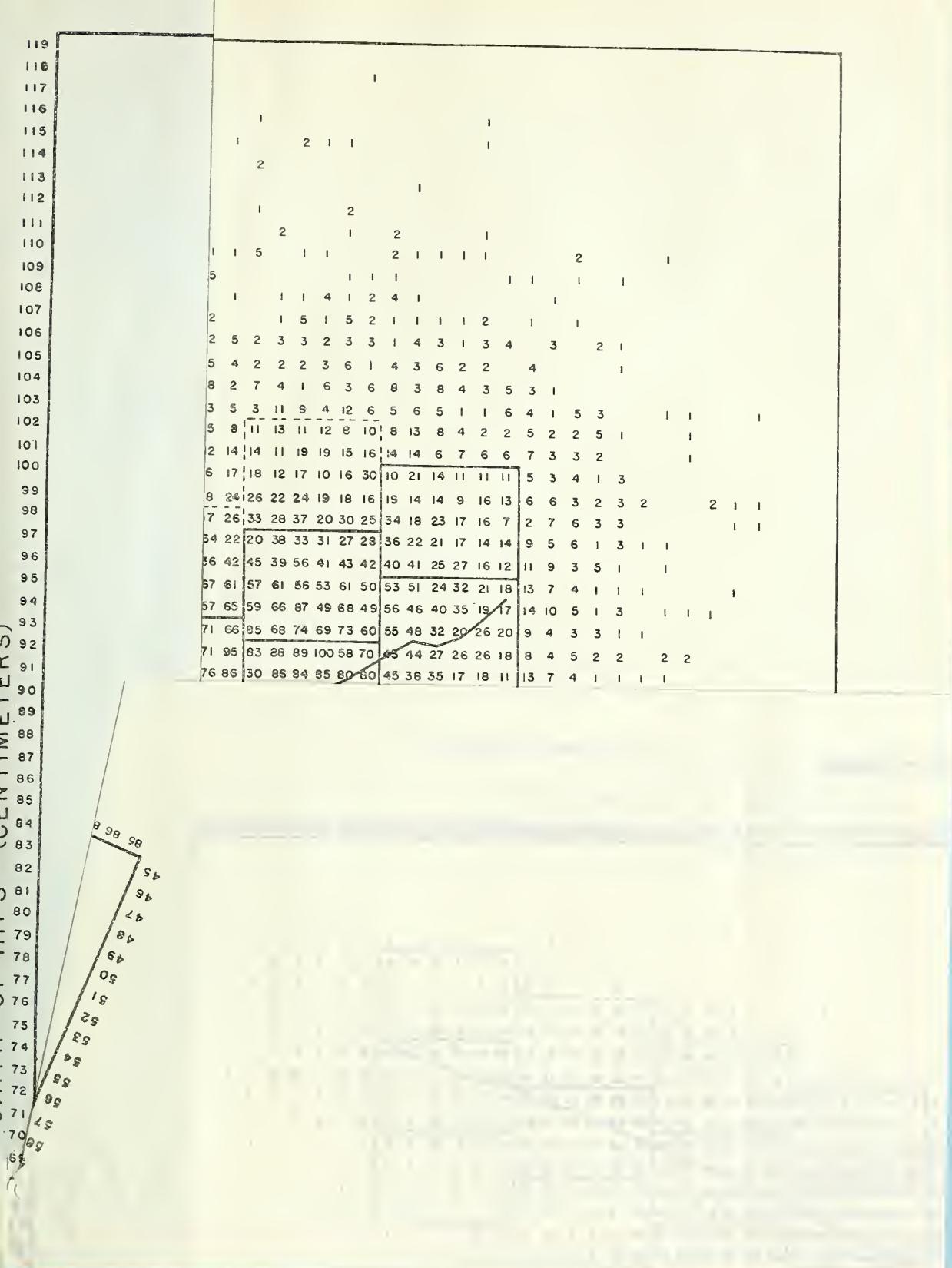
36. *Trunk girth, vertical.*—The subject is asked to pull up the measuring garment by the waist band. His position is his normal erect posture. The feet are placed slightly apart so that the tape can pass freely between the thighs. The observer is in front of the subject with eyes at the level of the nipples. With younger subjects it is convenient to sit down. About 1 m. of tape is drawn from the case. The zero end of the tape is passed midway between the landmarks of the neck base and the armscye. In the back, the tape passes between the buttocks. In the front, at the crotch, the tape passes centrally over the genitals without constriction. Before the reading is made, the subject is asked to relax his shoulders to their normal rest position. The subject breathes normally. The middle value between the largest and the smallest reading is recorded.



GIRTH OF HIPS (CENTIMETERS)

STATURE (CENTIMETERS)

FIGURE 1.—Distribution of boys 4 to 17 years of age on the basis of stature and girth of hips.



GIRTH OF HIPS (CENTIMETERS)

STATURE (CENTIMETERS)

FIGURE 2.—Distribution of girls 4 to 17 years of age on the basis of stature and girth of hips.

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